

# Activity No. 1: Number System Converter and Arithmetic Calculator

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## System Requirements

1. Hardware: A computer or laptop.
2. Software: A modern web browser like Google Chrome or Microsoft Edge.
3. Network: An active internet connection to load the Tailwind CSS framework via CDN and Google Fonts.
4. Tools: A basic text editor such as Notepad for code modification.

## Algorithm

1. Start the program.
2. Initialize the user interface with three default input cases, a custom arithmetic expression field, and a dedicated complement subtraction tool.
3. Dynamically assign a sequential alphabetical variable (A, B, C...) to each input case.
4. Prompt the user to enter a number and select its corresponding base (Binary, Octal, Decimal, Hexadecimal) for each active input.
5. Validate each input string against the allowed characters for its selected base, including support for fractional floating-point values.
6. Calculate the standard conversions (Binary, Octal, Decimal, Hexadecimal).
7. Calculate the 1's Complement and 2's Complement for the integer part of each valid input using dynamic bit-padding (8-bit, 16-bit, 32-bit depending on magnitude).
8. Display all six representations (Binary, Octal, Decimal, Hexadecimal, 1's Comp, 2's Comp) in the individual results grid.
9. Convert all valid input strings into a standard decimal floating-point format using a custom fractional parser and map them to their assigned alphabetical variables.
10. Read the custom arithmetic expression input by the user. Parse the expression using the Shunting-yard algorithm to evaluate strict operator precedence and parenthetical grouping logic.
11. Display the formatted equation breakdown and final computed arithmetic result in all base systems and complement formats.
12. Populate the Complement Subtraction Tool with valid variables. When two variables are selected (Minuend and Subtrahend), compute and display the step-by-step binary subtraction process utilizing both the 1's Complement Method (End-Around Carry) and 2's Complement Method.
13. Populate the BCD Addition & Subtraction Tool. When two variables and an operation (+ or -) are selected, compute and display the BCD sum (with +6 correction if needed) or the difference using 9's and 10's complement subtraction.
14. Automatically recalculate and reassign variables (preventing alphabetical gaps) if the user dynamically adds or removes input cases.
15. Handle and display explicit errors for invalid inputs, syntax errors, mismatched parentheses, or mathematical impossibilities (division by zero).

16. End the program.

## Pseudocode

```
START PROGRAM
  SET caseCounter = 0
  CALL addCase() THREE TIMES to render initial interface

  FUNCTION getComplements(num)
    SET intVal = TRUNCATE(num)
    SET bits = CALCULATE required bits (8, 16, 32) based on intVal
    SET onesComp = INVERT BITS OF (absolute intVal in binary)
    SET twosComp = ADD 1 TO onesComp
    RETURN onesComp, twosComp
  END FUNCTION

  FUNCTION calculateTotal()
    READ exprString FROM math-expression input
    SET variables = EMPTY MAP

    FOR EACH active case container:
      READ rawValue AND inBase
      IF rawValue IS INVALID THEN ABORT AND DISPLAY ERROR

      SET decValue = CUSTOM PARSE rawValue TO Decimal FLOAT
      STORE decValue IN variables[assignedLetter]

      DISPLAY Base Conversions AND getComplements(decValue)
    END FOR

    POPULATE Complement Subtraction Selectors WITH keys OF variables
    POPULATE BCD Arithmetic Selectors WITH keys OF variables

    TRY
      SET postfix = SHUNTING_YARD_PARSE(exprString)
      SET finalResult = EVALUATE_POSTFIX(postfix, variables)

      DISPLAY finalResult IN Base 2, 8, 10, 16, 1's Comp, 2's Comp
    CATCH ERROR
      DISPLAY explicit error message
    END TRY
  END FUNCTION

  FUNCTION calcComplementSub()
    READ minuendVar AND subtrahendVar
    SET M = TRUNCATE(variables[minuendVar])
    SET S = ABS(TRUNCATE(variables[subtrahendVar]))

    SET paddedM = BINARY(M) padded to common bit length
    SET paddedS = BINARY(S) padded to common bit length

    // 1's Complement Method
```

```
SET onesS = INVERT(paddedS)
SET sum1 = paddedM + onesS
IF carry THEN sum1 = sum1 + 1 (End Around Carry)

// 2's Complement Method
SET twosS = INVERT(paddedS) + 1
SET sum2 = paddedM + twosS (Discard Carry)

DISPLAY step-by-step calculations FOR both methods
END FUNCTION

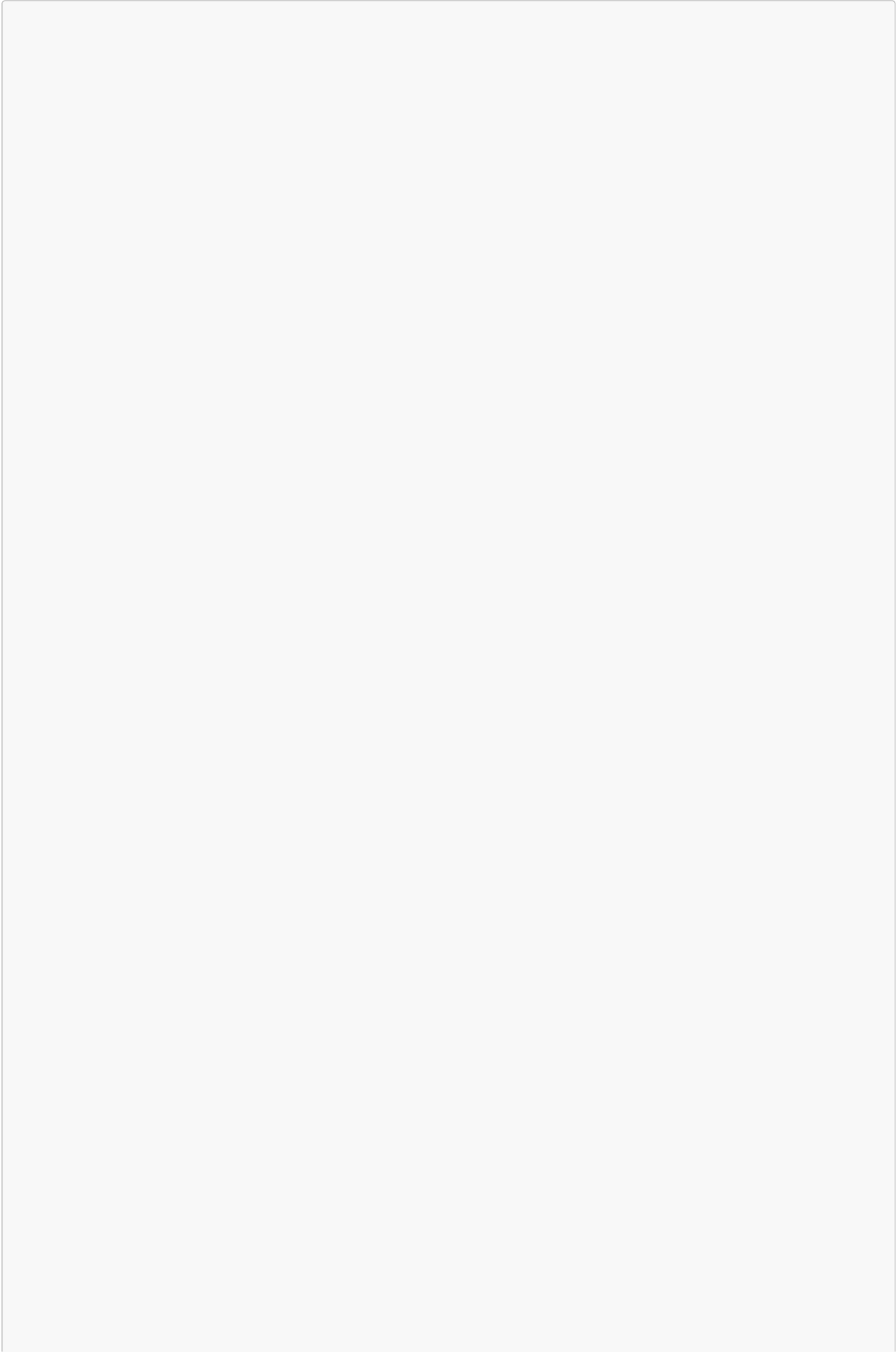
FUNCTION calcBCD()
  READ var1, var2, AND operator (+ or -)
  SET val1 = ABSOLUTE(TRUNCATE(variables[var1]))
  SET val2 = ABSOLUTE(TRUNCATE(variables[var2]))

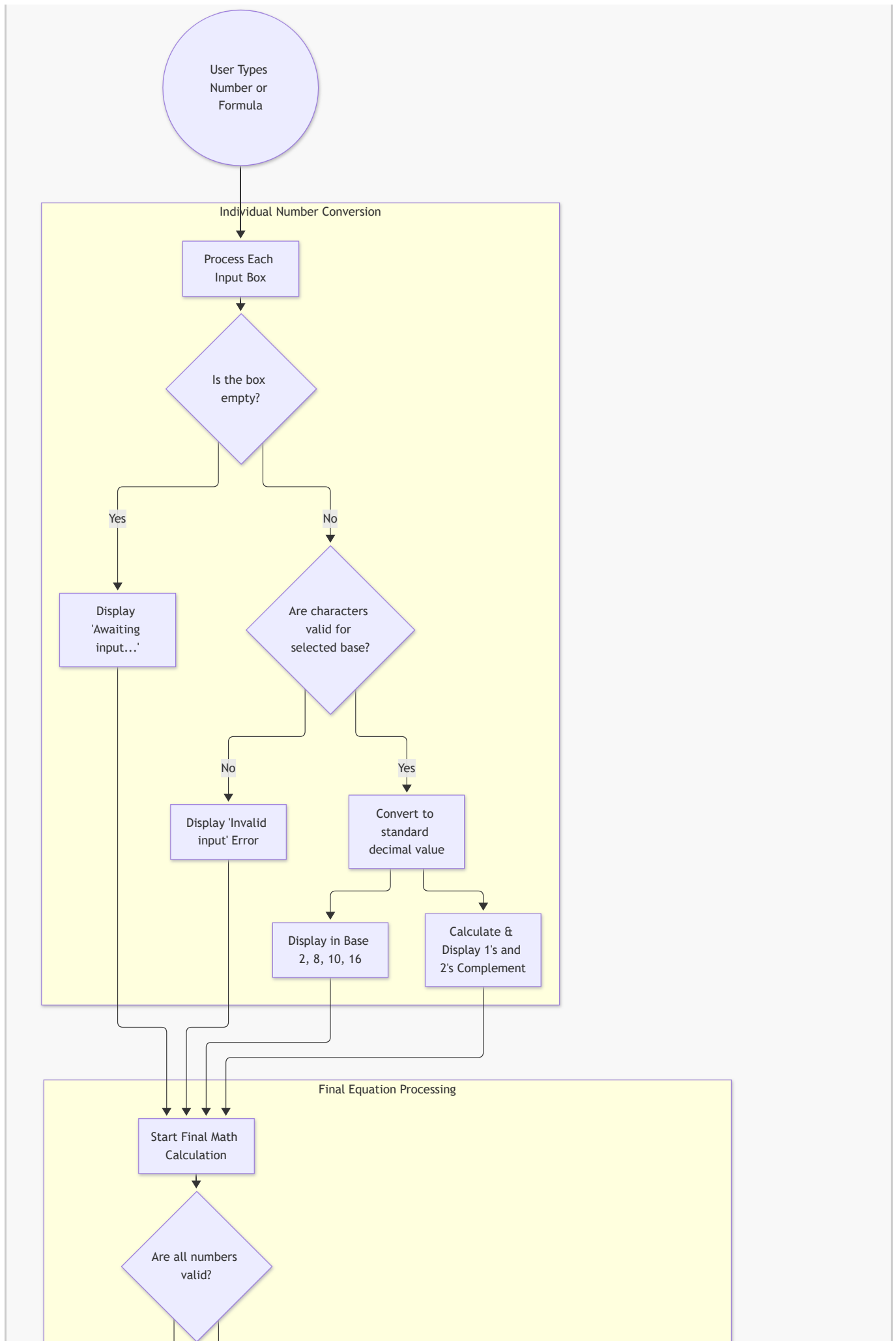
  IF operator IS '+' THEN
    COMPUTE Sum
    IF any 4-bit chunk > 9 OR generates carry THEN
      ADD 0110 (6) to correct it
      DISPLAY BCD Sum and correction rules applied
    ELSE
      COMPUTE 9's complement of val2 (subtract from 9)
      COMPUTE 10's complement of val2 (9's comp + 1)

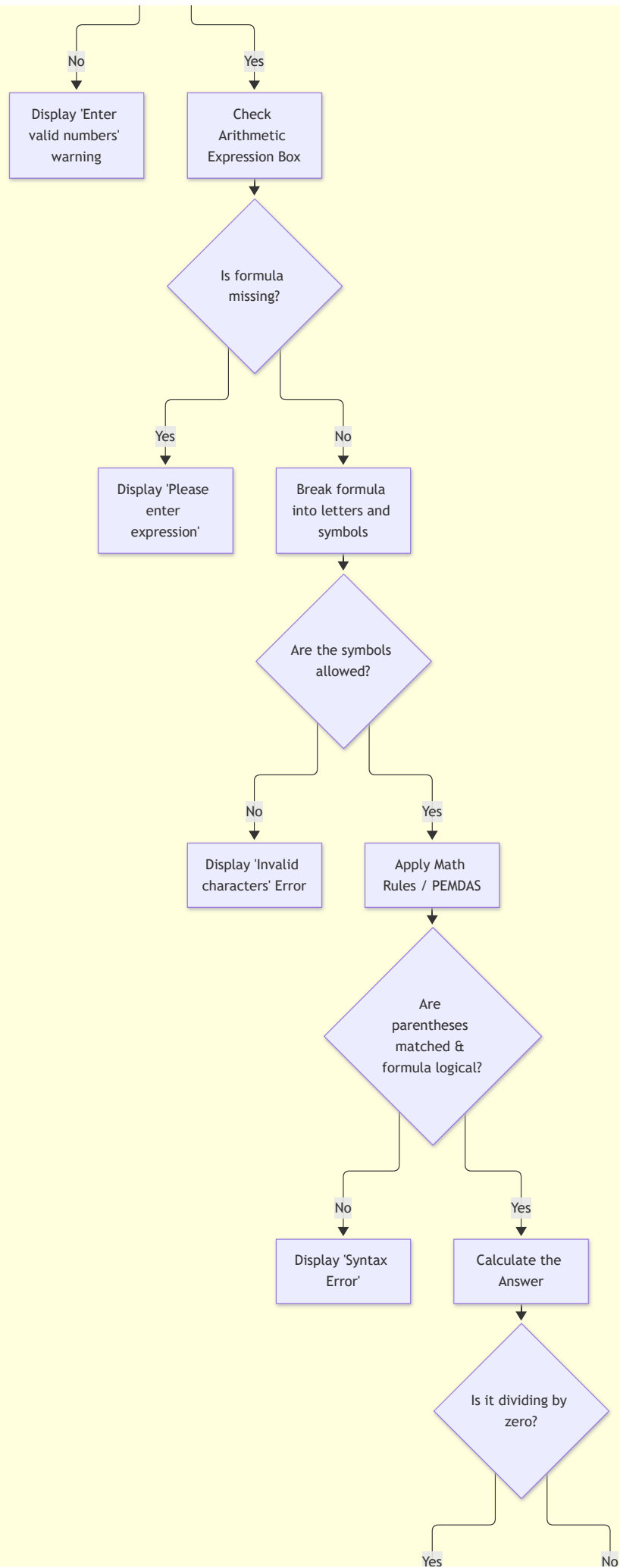
      ADD 9's comp to val1 -> Handle End-Around Carry or Negative
      ADD 10's comp to val1 -> Discard Carry or Negative

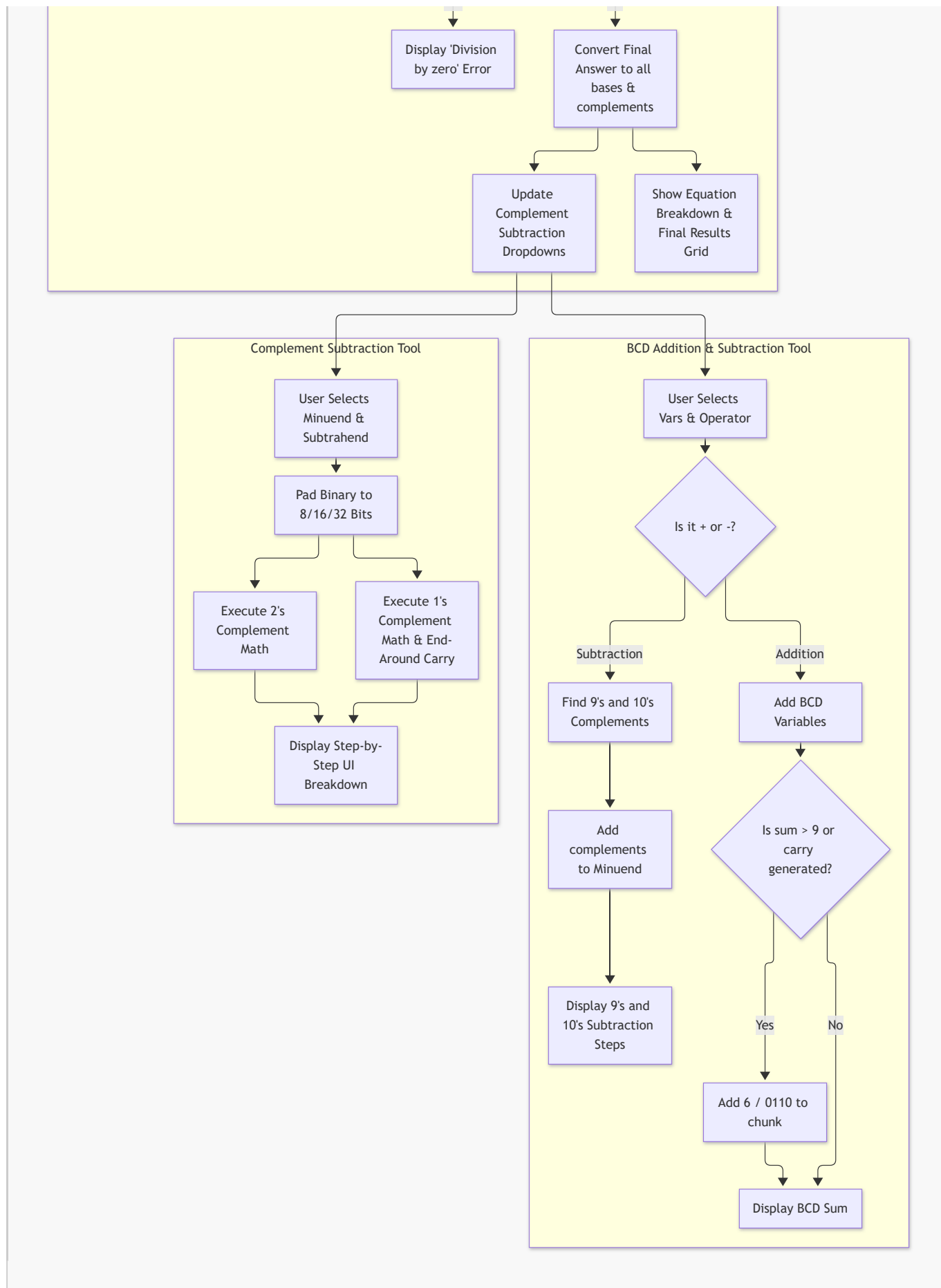
      DISPLAY 9's and 10's complement subtraction steps
    END IF
  END FUNCTION
END PROGRAM
```

## Flowchart









## Program Implementation

```

<!DOCTYPE html>
<html lang="en" class="dark">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Number System Converter & Calculator</title>

  <link rel="preconnect" href="https://fonts.googleapis.com"
(https://fonts.googleapis.com)">
  <link rel="preconnect" href="https://fonts.gstatic.com"
(https://fonts.gstatic.com)" crossorigin>
  <link href="https://fonts.googleapis.com/css2?
family=Plus+Jakarta+Sans:wght@400;600;800&family=Space+Grotesk:wght@500;700&displa
y=swap"(https://fonts.googleapis.com/css2?
family=Plus+Jakarta+Sans:wght@400;600;800&family=Space+Grotesk:wght@500;700&displa
y=swap)" rel="stylesheet">

  <script src="https://cdn.tailwindcss.com"(https://cdn.tailwindcss.com)">
</script>
  <script>
    tailwind.config = {
      darkMode: 'class',
      theme: {
        extend: {
          fontFamily: {
            sans: ['Plus Jakarta Sans', 'sans-serif'],
            mono: ['Space Grotesk', 'monospace'],
          }
        }
      }
    }
  </script>
</head>
<body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-
slate-800 dark:text-indigo-100 transition-colors duration-300">

  <div class="max-w-6xl mx-auto">
    <div class="flex justify-end mb-4">
      <button onclick="toggleTheme()" class="p-3 rounded-full shadow-lg bg-indigo-
100 dark:bg-slate-800 text-indigo-700 dark:text-purple-400 hover:bg-indigo-200
dark:hover:bg-slate-700 border border-indigo-200 dark:border-indigo-800
transition-colors focus:outline-none focus:ring-2 focus:ring-indigo-500"
title="Toggle Light/Dark Mode">
        <svg class="w-6 h-6 hidden dark:block" fill="none" stroke="currentColor"
viewBox="0 0 24 24" xmlns="http://www.w3.org/2000/svg"
(http://www.w3.org/2000/svg)">
          <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2"
d="M12 3v1m0 16v1m9-9h-1M4 12H3m15.364 6.364l-.707-.707M6.343
6.343l-.707-.707m12.728 0l-.707.707M6.343 17.657l-.707.707M16 12a4 4 0 11-8 0 4 4
0 018 0z"></path>
        </svg>
        <svg class="w-6 h-6 block dark:hidden" fill="none" stroke="currentColor"

```

```

viewBox="0 0 24 24" xmlns="http://www.w3.org/2000/svg"
(http://www.w3.org/2000/svg)">
    <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2"
d="M20.354 15.354A9 9 0 018.646 3.646 9.003 9.003 0 0012 21a9.003 9.003 0 008.354-
5.646z"></path>
</svg>
</button>
</div>

<div class="bg-white dark:bg-slate-900 p-6 md:p-8 rounded-2xl shadow-xl border
border-slate-200 dark:border-indigo-900 transition-colors duration-300">
    <h1 class="text-3xl md:text-4xl font-extrabold mb-8 text-center text-indigo-
700 dark:text-purple-400 font-mono tracking-tight">System Converter &
Calculator</h1>

    <div class="mb-8 p-6 bg-indigo-50 dark:bg-indigo-950/30 border border-
indigo-200 dark:border-indigo-800 rounded-xl">
        <label class="block text-lg font-bold text-indigo-800 dark:text-purple-300
mb-3 text-center">Arithmetic Expression</label>
        <p class="text-sm text-center text-indigo-600 dark:text-indigo-400 mb-
4">Use variables (A, B, C...) and operators (+, -, *, /, parentheses). Example:
<strong>(A + B - C) * D</strong></p>
        <input type="text" id="math-expression" class="w-full md:w-3/4 mx-auto
block bg-white dark:bg-slate-900 border border-indigo-300 dark:border-indigo-700
p-4 rounded-lg text-2xl text-slate-900 dark:text-white font-bold text-center
focus:outline-none focus:ring-4 focus:ring-indigo-500 dark:focus:ring-purple-500
transition-colors shadow-sm uppercase tracking-widest" placeholder="A + B - C"
value="A + B - C" oninput="calculateTotal()">
    </div>

    <div id="cases-container" class="grid grid-cols-1 gap-6">
</div>

    <div class="mt-6 text-center">
        <button onclick="addCase()" class="bg-indigo-600 hover:bg-indigo-700
dark:bg-slate-800 dark:hover:bg-slate-700 border border-transparent dark:border-
indigo-700 text-white dark:text-purple-300 font-bold py-3 px-8 rounded-xl
transition-colors shadow-lg focus:outline-none focus:ring-2 focus:ring-indigo-
500">
            + Add Input Number
        </button>
    </div>

    <div class="mt-12 p-8 bg-slate-900 dark:bg-black border-2 border-indigo-500
dark:border-purple-600 rounded-2xl shadow-2xl relative overflow-hidden">
        <div class="absolute top-0 left-0 w-full h-1 bg-gradient-to-r from-indigo-
500 via-purple-500 to-pink-500"></div>
        <h2 class="text-2xl font-bold text-white mb-6 text-center">Final
Arithmetic Result</h2>
        <div id="final-total-container" class="text-center">
            <p class="text-slate-400 text-lg">Enter valid numbers and a proper
expression above to compute the result.</p>
        </div>
    </div>

```

```

<div class="mt-8 p-8 bg-slate-800 dark:bg-slate-950 border-2 border-emerald-500 dark:border-emerald-600 rounded-2xl shadow-xl relative overflow-hidden">
  <div class="absolute top-0 left-0 w-full h-1 bg-gradient-to-r from-emerald-400 to-teal-500"></div>
  <h2 class="text-2xl font-bold text-white mb-2 text-center">Complement Subtraction Method</h2>
  <p class="text-sm text-center text-emerald-400 mb-6">Select two variables to demonstrate step-by-step subtraction (Minuend - Subtrahend) using complements. Uses absolute integer parts.</p>

  <div class="flex justify-center items-center gap-4 mb-6">
    <select id="comp-minuend" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-emerald-500 font-bold text-lg"
      onchange="calcComplementSub()"></select>
    <span class="text-white text-3xl font-bold">-</span>
    <select id="comp-subtrahend" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-emerald-500 font-bold text-lg"
      onchange="calcComplementSub()"></select>
  </div>

  <div id="comp-sub-output" class="text-slate-300 text-center font-mono">
    Awaiting variable selection...
  </div>
</div>

<div class="mt-8 p-8 bg-slate-800 dark:bg-slate-950 border-2 border-blue-500 dark:border-blue-600 rounded-2xl shadow-xl relative overflow-hidden">
  <div class="absolute top-0 left-0 w-full h-1 bg-gradient-to-r from-blue-400 to-cyan-500"></div>
  <h2 class="text-2xl font-bold text-white mb-2 text-center">BCD Addition & Subtraction</h2>
  <p class="text-sm text-center text-blue-400 mb-6">Select two variables and an operation. Subtraction displays both 9's and 10's complement methods.</p>

  <div class="flex justify-center items-center gap-4 mb-6">
    <select id="bcd-var1" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-blue-500 font-bold text-lg"
      onchange="calcBCD()"></select>
    <select id="bcd-op" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-blue-500 font-bold text-lg"
      onchange="calcBCD()">
      <option value="+">+</option>
      <option value="-">-</option>
    </select>
    <select id="bcd-var2" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-blue-500 font-bold text-lg"
      onchange="calcBCD()"></select>
  </div>

  <div id="bcd-output" class="text-slate-300 text-center font-mono">
    Awaiting variable selection...
  </div>
</div>

```

```
</div>
</div>

<script>
  let caseCounter = 0;
  window.globalVars = {};

  function toggleTheme() {
    document.documentElement.classList.toggle('dark');
  }

  function parseToDecimal(str, base) {
    const isNegative = str.startsWith('-');
    const cleanStr = str.replace('-', '');
    const parts = cleanStr.split('.');

    let intPart = parseInt(parts[0] || '0', base);
    let fracPart = 0;

    if (parts.length > 1) {
      let fracStr = parts[1];
      let divisor = base;
      for (let i = 0; i < fracStr.length; i++) {
        fracPart += parseInt(fracStr[i], base) / divisor;
        divisor *= base;
      }
    }

    let total = intPart + fracPart;
    return isNegative ? -total : total;
  }

  function formatBase(num, base) {
    if (isNaN(num)) return "NaN";
    const isNeg = num < 0;
    const absVal = Math.abs(num);
    let str = absVal.toString(base).toUpperCase();
    return (isNeg ? '-' : '') + str;
  }

  function getComplements(num) {
    if (isNaN(num)) return { ones: "NaN", twos: "NaN" };
    let intVal = Math.trunc(Math.abs(num));
    let binStr = intVal.toString(2);

    let bits = 8;
    while (binStr.length > bits - 1) bits += 8;
    binStr = binStr.padStart(bits, '0');

    let onesComp = binStr.split('').map(b => b === '0' ? '1' : '0').join('');

    let twosComp = '';
    let carry = 1;
```

```

    for (let i = bits - 1; i >= 0; i--) {
      let sum = parseInt(onesComp[i]) + carry;
      twosComp = (sum % 2) + twosComp;
      carry = Math.floor(sum / 2);
    }

    return { ones: onesComp, twos: twosComp, bits: bits };
  }

function addBinaryStr(a, b) {
  let res = '', carry = 0;
  for(let i = a.length - 1; i >= 0; i--) {
    let sum = parseInt(a[i]) + parseInt(b[i]) + carry;
    res = (sum % 2) + res;
    carry = Math.floor(sum / 2);
  }
  return { sum: res, carry: carry };
}

function createCaseHTML(id) {
  return `
    <div id="case-${id}" data-var="" class="p-6 border border-slate-200
dark:border-indigo-800 rounded-xl bg-slate-50 dark:bg-slate-800 relative shadow-sm
dark:shadow-inner transition-colors duration-300">
      <div class="flex justify-between items-center mb-4">
        <label class="case-label font-bold text-xl text-indigo-800 dark:text-
purple-300 bg-indigo-100 dark:bg-indigo-900/50 px-4 py-1 rounded-full border
border-indigo-200 dark:border-indigo-700">Input</label>
        <button onclick="removeCase(${id})" class="remove-btn p-2 text-red-500
dark:text-red-400 hover:text-red-700 dark: hover:text-red-300 hover:bg-red-100
dark: hover:bg-red-900/30 rounded-lg hidden transition-colors focus:outline-none"
title="Remove Input">
          <svg class="w-5 h-5" fill="none" stroke="currentColor" viewBox="0 0
24 24" xmlns="http://www.w3.org/2000/svg"(http://www.w3.org/2000/svg)">
            <path stroke-linecap="round" stroke-linejoin="round" stroke-
width="2" d="M6 18L18 6M6 6l12 12"></path>
          </svg>
        </button>
      </div>

      <div class="grid grid-cols-1 md:grid-cols-2 gap-4 mb-4">
        <div>
          <label class="block text-sm font-semibold text-slate-600 dark:text-
indigo-300 mb-1">Value</label>
          <input type="text" id="val-${id}" class="w-full bg-white dark:bg-
slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg
font-mono text-slate-900 dark:text-white focus:outline-none focus:ring-2
focus:ring-indigo-500 dark:focus:ring-purple-500 placeholder-slate-400
dark:placeholder-slate-500 transition-colors" placeholder="0"
oninput="handleInput(${id})">
        </div>
        <div>
          <label class="block text-sm font-semibold text-slate-600 dark:text-
indigo-300 mb-1">Base System</label>

```

```

        <select id="inBase-${id}" class="w-full bg-white dark:bg-slate-900
border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg text-slate-
900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500
dark:focus:ring-purple-500 transition-colors" onchange="handleInput(${id})">
            <option value="2">Binary (Base 2)</option>
            <option value="8">Octal (Base 8)</option>
            <option value="10" selected>Decimal (Base 10)</option>
            <option value="16">Hexadecimal (Base 16)</option>
        </select>
    </div>
</div>

    <div id="out-${id}" class="mt-4 p-4 bg-white dark:bg-slate-950 border
border-slate-200 dark:border-indigo-900 rounded-lg min-h-[80px] flex items-center
justify-center text-slate-500 dark:text-indigo-400 transition-colors duration-
300">
        Awaiting input...
    </div>
</div>
`;
}

function reindexLabels() {
    const cases = document.querySelectorAll('[id^="case-"]');
    cases.forEach((caseEl, index) => {
        const varLetter = String.fromCharCode(65 + index);
        caseEl.setAttribute('data-var', varLetter);
        const labelEl = caseEl.querySelector('.case-label');
        if (labelEl) {
            labelEl.textContent = `Input ${varLetter}`;
        }
    });
}

function addCase() {
    caseCounter++;
    const container = document.getElementById('cases-container');
    container.insertAdjacentHTML('beforeend', createCaseHTML(caseCounter));
    reindexLabels();
    updateRemoveButtons();
    calculateTotal();
}

function removeCase(id) {
    const caseEl = document.getElementById(`case-${id}`);
    if (caseEl && document.querySelectorAll('[id^="case-"]').length > 2) {
        caseEl.remove();
        reindexLabels();
        updateRemoveButtons();
        calculateTotal();
    }
}

function updateRemoveButtons() {

```

```

const buttons = document.querySelectorAll('.remove-btn');
if (buttons.length <= 2) {
  buttons.forEach(btn => btn.classList.add('hidden'));
} else {
  buttons.forEach(btn => btn.classList.remove('hidden'));
}
}

function handleInput(id) {
  convertIndividualNumber(id);
  calculateTotal();
}

function convertIndividualNumber(id) {
  const rawValue = document.getElementById(`val-${id}`).value.trim();
  const inBase = parseInt(document.getElementById(`inBase-${id}`).value);
  const outputDiv = document.getElementById(`out-${id}`);

  if (rawValue === "") {
    outputDiv.innerHTML = "Awaiting input...";
    outputDiv.className = "mt-4 p-4 bg-white dark:bg-slate-950 border border-slate-200 dark:border-indigo-900 rounded-lg min-h-[80px] flex items-center justify-center text-slate-500 dark:text-indigo-400 text-lg transition-colors";
    return;
  }

  let isValid = false;
  if (inBase === 2) isValid = /^-?[01]+(\.[01]+)?$/ .test(rawValue);
  if (inBase === 8) isValid = /^-?[0-7]+(\.[0-7]+)?$/ .test(rawValue);
  if (inBase === 10) isValid = /^-?[0-9]+(\.[0-9]+)?$/ .test(rawValue);
  if (inBase === 16) isValid = /^-?[0-9a-fA-F]+(\.[0-9a-fA-F]+)?$/ .test(rawValue);

  if (!isValid) {
    outputDiv.innerHTML = "<span class='text-red-600 dark:text-red-400 font-bold'>Invalid input for selected base.</span>";
    outputDiv.className = "mt-4 p-4 bg-white dark:bg-slate-950 border border-slate-200 dark:border-indigo-900 rounded-lg min-h-[80px] flex items-center justify-center transition-colors";
    return;
  }

  let decValue = parseToDecimal(rawValue, inBase);
  let comps = getComplements(decValue);

  const binStr = formatBase(decValue, 2);
  const octStr = formatBase(decValue, 8);
  const decStr = formatBase(decValue, 10);
  const hexStr = formatBase(decValue, 16);

  outputDiv.innerHTML = `
    <div class="grid grid-cols-2 lg:grid-cols-3 xl:grid-cols-6 gap-3 w-full">
      <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">

```

```

        <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400
mb-1">Binary</span>
        <div class="overflow-x-auto"><span class="text-sm font-mono font-bold
text-indigo-700 dark:text-purple-300 break-all">${binStr}</span></div>
    </div>
    <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-
slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
        <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400
mb-1">Octal</span>
        <div class="overflow-x-auto"><span class="text-sm font-mono font-bold
text-indigo-700 dark:text-purple-300 break-all">${octStr}</span></div>
    </div>
    <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-
slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
        <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400
mb-1">Decimal</span>
        <div class="overflow-x-auto"><span class="text-sm font-mono font-bold
text-indigo-700 dark:text-purple-300 break-all">${decStr}</span></div>
    </div>
    <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-
slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
        <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400
mb-1">Hexadecimal</span>
        <div class="overflow-x-auto"><span class="text-sm font-mono font-bold
text-indigo-700 dark:text-purple-300 break-all">${hexStr}</span></div>
    </div>
    <div class="bg-indigo-50 dark:bg-indigo-950/40 p-3 rounded-lg border
border-indigo-200 dark:border-indigo-800 min-w-0 flex flex-col">
        <span class="text-xs font-semibold text-indigo-600 dark:text-indigo-
400 mb-1">1's Comp (Int)</span>
        <div class="overflow-x-auto"><span class="text-sm font-mono font-bold
text-indigo-800 dark:text-purple-200 break-all">${comps.ones}</span></div>
    </div>
    <div class="bg-indigo-50 dark:bg-indigo-950/40 p-3 rounded-lg border
border-indigo-200 dark:border-indigo-800 min-w-0 flex flex-col">
        <span class="text-xs font-semibold text-indigo-600 dark:text-indigo-
400 mb-1">2's Comp (Int)</span>
        <div class="overflow-x-auto"><span class="text-sm font-mono font-bold
text-indigo-800 dark:text-purple-200 break-all">${comps.twos}</span></div>
    </div>
</div>
`;
    outputDiv.className = "mt-4 p-3 bg-white dark:bg-slate-950 border border-
slate-200 dark:border-indigo-900 rounded-lg min-h-[80px] transition-colors";
}

function evaluatePostfix(postfix, variables) {
    const valStack = [];
    for (let token of postfix) {
        if (typeof token === 'number') {
            valStack.push(token);
        } else if (variables.hasOwnProperty(token)) {
            valStack.push(variables[token]);
        } else {

```

```

        if (valStack.length < 2) throw new Error("Invalid expression syntax.");
        const b = valStack.pop();
        const a = valStack.pop();
        if (token === '+') valStack.push(a + b);
        else if (token === '-') valStack.push(a - b);
        else if (token === '*') valStack.push(a * b);
        else if (token === '/') {
            if (b === 0) throw new Error("Division by zero error.");
            valStack.push(a / b);
        }
    }
}
if (valStack.length !== 1) throw new Error("Invalid expression syntax.");
return valStack[0];
}

function calculateTotal() {
    const cases = document.querySelectorAll('[id^="case-"]');
    const totalDiv = document.getElementById('final-total-container');
    const exprString = document.getElementById('math-expression').value.trim().toUpperCase();

    window.globalVars = {};
    let displayStrs = {};
    let allValid = true;

    cases.forEach(caseEl => {
        const id = caseEl.id.split('-')[1];
        const varName = caseEl.getAttribute('data-var');
        const rawValue = document.getElementById(`val-${id}`).value.trim();
        const inBase = parseInt(document.getElementById(`inBase-${id}`).value);

        if (!rawValue) {
            allValid = false;
            return;
        }

        let isValid = false;
        if (inBase === 2) isValid = /^-?[01]+(\.[01]+)?$/ .test(rawValue);
        if (inBase === 8) isValid = /^-?[0-7]+(\.[0-7]+)?$/ .test(rawValue);
        if (inBase === 10) isValid = /^-?[0-9]+(\.[0-9]+)?$/ .test(rawValue);
        if (inBase === 16) isValid = /^-?[0-9a-fA-F]+(\.[0-9a-fA-F]+)?$/ .test(rawValue);

        if (!isValid) {
            allValid = false;
            return;
        }

        let decValue = parseToDecimal(rawValue, inBase);
        window.globalVars[varName] = decValue;

        const sub = inBase === 2 ? '2' : inBase === 8 ? '8' : inBase === 10 ? '10' : '16';

```

```

    displayStrs[varName] = `(${rawValue})${sub}`;
  });

  updateComplementSelects();
  updateBCDSelects();

  if (!allValid || Object.keys(window.globalVars).length === 0) {
    totalDiv.innerHTML = "<p class='text-slate-400 text-lg'>Enter valid
numbers in all active fields to compute.</p>";
    return;
  }

  if (!exprString) {
    totalDiv.innerHTML = "<p class='text-slate-400 text-lg'>Please enter a
mathematical expression.</p>";
    return;
  }

  const tokens = exprString.match(/[A-Z]+|[0-9]*\.[0-9]+|[+\-*/()]/g);
  if (!tokens) {
    totalDiv.innerHTML = "<p class='text-red-400 text-xl font-bold'>Error:
Invalid characters in expression.</p>";
    return;
  }

  const precedence = { '+': 1, '-': 1, '*': 2, '/': 2 };
  const postfix = [];
  const opStack = [];
  let formattedEqTokens = [];

  try {
    for (let token of tokens) {
      if (window.globalVars.hasOwnProperty(token)) {
        postfix.push(token);
        formattedEqTokens.push(`<span class="text-white">${displayStrs[token]}
</span>`);
      } else if (/^[0-9]*\.[0-9]+$/.test(token)) {
        postfix.push(Number(token));
        formattedEqTokens.push(`<span class="text-white">${token}</span>`);
      } else if (['+', '-', '*', '/'].includes(token)) {
        let symbol = token === '*' ? 'x' : token === '/' ? '÷' : token;
        formattedEqTokens.push(`<span class="text-pink-400 mx-1">${symbol}
</span>`);
        while (opStack.length > 0 && opStack[opStack.length - 1] !== '(' &&
          precedence[opStack[opStack.length - 1]] >= precedence[token]) {
          postfix.push(opStack.pop());
        }
        opStack.push(token);
      } else if (token === '(') {
        formattedEqTokens.push(`<span class="text-indigo-400">(</span>`);
        opStack.push(token);
      } else if (token === ')') {
        formattedEqTokens.push(`<span class="text-indigo-400"></span>`);
        while (opStack.length > 0 && opStack[opStack.length - 1] !== '(') {

```

```

        postfix.push(opStack.pop());
      }
      if (opStack.length === 0) throw new Error("Mismatched parentheses.");
      opStack.pop();
    } else {
      throw new Error(`Unknown variable: ${token}`);
    }
  }
  while (opStack.length > 0) {
    const op = opStack.pop();
    if (op === '(' || op === ')') throw new Error("Mismatched
parentheses.");
    postfix.push(op);
  }

  const finalResult = evaluatePostfix(postfix, window.globalVars);
  const comps = getComplements(finalResult);

  const binStr = formatBase(finalResult, 2);
  const octStr = formatBase(finalResult, 8);
  const decStr = formatBase(finalResult, 10);
  const hexStr = formatBase(finalResult, 16);

  totalDiv.innerHTML = `
    <div class="mb-6 p-4 bg-slate-800 rounded-lg border border-slate-700
shadow-inner overflow-x-auto">
      <p class="text-indigo-300 text-sm font-bold uppercase tracking-wider
mb-2">Equation Breakdown</p>
      <p class="text-xl md:text-2xl font-mono whitespace-
nowrap">${formattedEqTokens.join('')} <span class="text-pink-400 mx-2">=</span>
</p>
    </div>

    <div class="grid grid-cols-2 lg:grid-cols-3 xl:grid-cols-6 gap-4">
      <div class="bg-slate-800 p-4 rounded-xl border border-slate-700 text-
left overflow-hidden">
        <span class="block text-sm font-bold text-slate-400 mb-
1">Binary</span>
        <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-
mono font-bold text-green-400 break-all">${binStr}</span></div>
      </div>
      <div class="bg-slate-800 p-4 rounded-xl border border-slate-700 text-
left overflow-hidden">
        <span class="block text-sm font-bold text-slate-400 mb-
1">Octal</span>
        <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-
mono font-bold text-blue-400 break-all">${octStr}</span></div>
      </div>
      <div class="bg-slate-800 p-4 rounded-xl border border-slate-700 text-
left overflow-hidden">
        <span class="block text-sm font-bold text-slate-400 mb-
1">Decimal</span>
        <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-
mono font-bold text-yellow-400 break-all">${decStr}</span></div>

```

```

        </div>
        <div class="bg-slate-800 p-4 rounded-xl border border-slate-700 text-left overflow-hidden">
            <span class="block text-sm font-bold text-slate-400 mb-1">Hexadecimal</span>
            <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-mono font-bold text-pink-400 break-all">${hexStr}</span></div>
        </div>
        <div class="bg-slate-800 p-4 rounded-xl border border-indigo-500/50 text-left overflow-hidden">
            <span class="block text-sm font-bold text-indigo-300 mb-1">1's Comp (Int)</span>
            <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-mono font-bold text-indigo-400 break-all">${comps.ones}</span></div>
        </div>
        <div class="bg-slate-800 p-4 rounded-xl border border-indigo-500/50 text-left overflow-hidden">
            <span class="block text-sm font-bold text-indigo-300 mb-1">2's Comp (Int)</span>
            <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-mono font-bold text-indigo-400 break-all">${comps.twos}</span></div>
        </div>
    </div>
    `;
    } catch (err) {
        totalDiv.innerHTML = `
```

```

let mKey = document.getElementById('comp-minuend').value;
let sKey = document.getElementById('comp-subtrahend').value;
let out = document.getElementById('comp-sub-output');

if(!mKey || !sKey || window.globalVars[mKey] === undefined ||
window.globalVars[sKey] === undefined) {
  out.innerHTML = "Awaiting valid variables for calculation...";
  return;
}

let mVal = Math.trunc(window.globalVars[mKey]);
let sVal = Math.abs(Math.trunc(window.globalVars[sKey]));

let maxAbs = Math.max(Math.abs(mVal), sVal);
let bits = 8;
while(maxAbs >= Math.pow(2, bits - 1)) bits += 8;

function toSignedBin(val, bits) {
  if (val >= 0) return val.toString(2).padStart(bits, '0');
  let posBin = Math.abs(val).toString(2).padStart(bits, '0');
  let flipped = posBin.split('').map(b => b==='0'? '1': '0').join('');
  return addBinaryStr(flipped, '1'.padStart(bits, '0')).sum;
}

let mBin = toSignedBin(mVal, bits);
let sBinAbs = toSignedBin(sVal, bits);

let sOnes = sBinAbs.split('').map(b => b==='0'? '1': '0').join('');
let sTwos = addBinaryStr(sOnes, '1'.padStart(bits, '0')).sum;

let add1 = addBinaryStr(mBin, sOnes);
let finalOnes = add1.sum;
let endCarryTxt = "No carry, result is negative (in 1's complement form).";
if (add1.carry) {
  finalOnes = addBinaryStr(add1.sum, '1'.padStart(bits, '0')).sum;
  endCarryTxt = "Carry = 1. Add End-Around Carry (+1).";
}

let add2 = addBinaryStr(mBin, sTwos);
let finalTwos = add2.sum;

out.innerHTML = `
  <div class="grid grid-cols-1 md:grid-cols-2 gap-8 text-left mt-4">
    <div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-
inner">
      <h3 class="text-teal-400 font-bold mb-4 border-b border-slate-700 pb-
2">1's Complement Method</h3>
      <p class="text-slate-400 mb-1">M: <span class="text-white">${mBin}
</span></p>
      <p class="text-slate-400 mb-1">S: <span class="text-white">${sBinAbs}
</span></p>
      <p class="text-slate-400 mb-3 border-b border-slate-700 pb-2">1's
Comp(S): <span class="text-pink-400">${sOnes}</span></p>
      <p class="text-slate-400 mb-1">Add M + 1's Comp(S):</p>

```

```

        <p class="text-white font-bold tracking-widest bg-slate-800 p-2
rounded mb-2">${mBin}<br>+${sOnes}<br>-----<br>${add1.carry ? '1 ' : '0
'}${add1.sum}</p>
        <p class="text-emerald-400 text-sm mb-2">${endCarryTxt}</p>
        <p class="text-white font-bold tracking-widest bg-slate-800 p-2
rounded">Final: ${finalOnes}</p>
    </div>

    <div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-
inner">
        <h3 class="text-teal-400 font-bold mb-4 border-b border-slate-700 pb-
2">2's Complement Method</h3>
        <p class="text-slate-400 mb-1">M: <span class="text-white">${mBin}
</span></p>
        <p class="text-slate-400 mb-1">S: <span class="text-white">${sBinAbs}
</span></p>
        <p class="text-slate-400 mb-3 border-b border-slate-700 pb-2">2's
Comp(S): <span class="text-pink-400">${sTwos}</span></p>
        <p class="text-slate-400 mb-1">Add M + 2's Comp(S):</p>
        <p class="text-white font-bold tracking-widest bg-slate-800 p-2
rounded mb-2">${mBin}<br>+${sTwos}<br>-----<br>${add2.carry ? '1 ' : '0
'}${add2.sum}</p>
        <p class="text-emerald-400 text-sm mb-2">Discard carry out.</p>
        <p class="text-white font-bold tracking-widest bg-slate-800 p-2
rounded">Final: ${finalTwos}</p>
    </div>
</div>
`;
}

function toBCD(str) {
    return str.split('').map(d => parseInt(d).toString(2).padStart(4,
'0')).join(' ');
}

function getNinesComp(str) {
    return str.split('').map(d => (9 - parseInt(d)).toString()).join('');
}

function updateBCDSelects() {
    const v1Sel = document.getElementById('bcd-var1');
    const v2Sel = document.getElementById('bcd-var2');
    const v1Val = v1Sel.value;
    const v2Val = v2Sel.value;

    v1Sel.innerHTML = '';
    v2Sel.innerHTML = '';

    Object.keys(window.globalVars).forEach(key => {
        v1Sel.add(new Option(`Var ${key}`, key));
        v2Sel.add(new Option(`Var ${key}`, key));
    });

    if(v1Val && window.globalVars[v1Val] !== undefined) v1Sel.value = v1Val;

```

```

    if(v2Val && window.globalVars[v2Val] !== undefined) v2Sel.value = v2Val;
    else if(Object.keys(window.globalVars).length > 1) v2Sel.selectedIndex = 1;

    calcBCD();
}

function calcBCD() {
    let k1 = document.getElementById('bcd-var1').value;
    let k2 = document.getElementById('bcd-var2').value;
    let op = document.getElementById('bcd-op').value;
    let out = document.getElementById('bcd-output');

    if(!k1 || !k2 || window.globalVars[k1] === undefined ||
window.globalVars[k2] === undefined) {
        out.innerHTML = "Awaiting valid variables for calculation...";
        return;
    }

    let val1 = Math.abs(Math.trunc(window.globalVars[k1])).toString();
    let val2 = Math.abs(Math.trunc(window.globalVars[k2])).toString();

    let maxLen = Math.max(val1.length, val2.length);
    val1 = val1.padStart(maxLen, '0');
    val2 = val2.padStart(maxLen, '0');

    let bcd1 = toBCD(val1);
    let bcd2 = toBCD(val2);

    let html = `

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```

```

let tens2 = (parseInt(nines2, 10) + 1).toString().padStart(maxLen, '0');
let diff9 = parseInt(val1, 10) + parseInt(nines2, 10);
let diff10 = parseInt(val1, 10) + parseInt(tens2, 10);
let isNeg = parseInt(val1, 10) < parseInt(val2, 10);

html += `

23 / 52


```

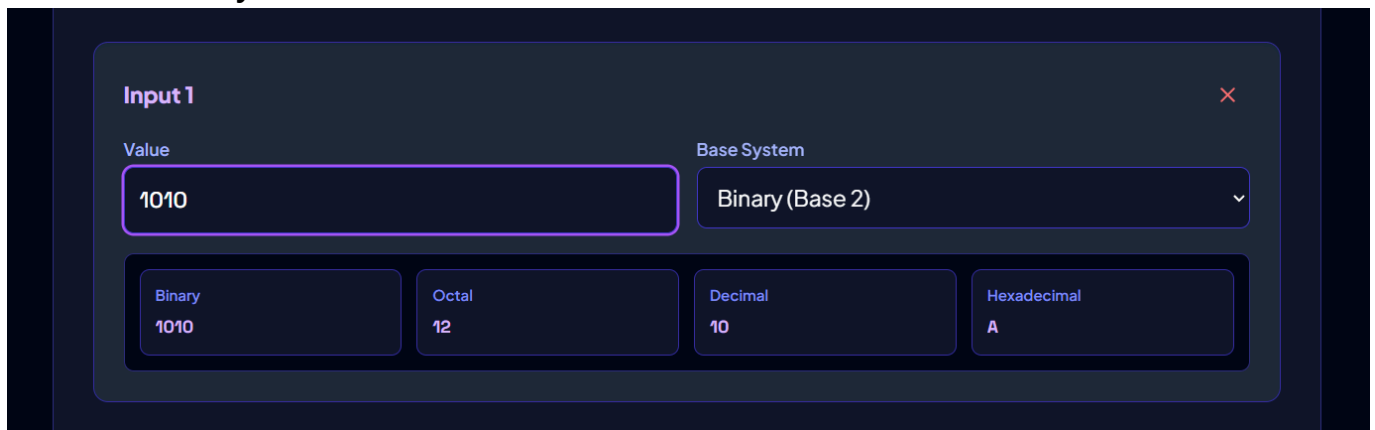
```
        out.innerHTML = html;
    }

    addCase();
    addCase();
    addCase();
</script>

</body>
</html>
```

## Test Cases

### Test Case 1: Binary Conversion



**Input 1** ✕

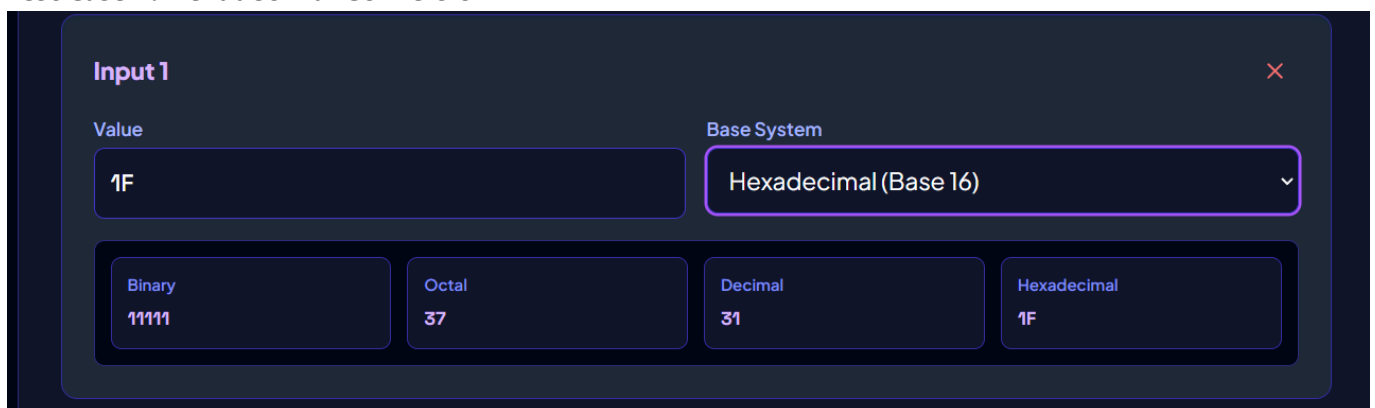
Value:

Base System: Binary (Base 2) ▾

|                |             |               |                  |
|----------------|-------------|---------------|------------------|
| Binary<br>1010 | Octal<br>12 | Decimal<br>10 | Hexadecimal<br>A |
|----------------|-------------|---------------|------------------|

1. Input Value: 1010
2. Selected Base: Binary (Base 2)
3. Expected Output: Binary: 1010, Octal: 12, Decimal: 10, Hexadecimal: A

### Test Case 2: Hexadecimal Conversion



**Input 1** ✕

Value:

Base System: Hexadecimal (Base 16) ▾

|                 |             |               |                   |
|-----------------|-------------|---------------|-------------------|
| Binary<br>11111 | Octal<br>37 | Decimal<br>31 | Hexadecimal<br>1F |
|-----------------|-------------|---------------|-------------------|

1. Input Value: 1F
2. Selected Base: Hexadecimal (Base 16)
3. Expected Output: Binary: 11111, Octal: 37, Decimal: 31, Hexadecimal: 1F

### Test Case 3: Invalid Input Handling

Input 1

Value

Base System

9

Octal (Base 8)

Invalid input for selected base.

1. Input Value: 9
2. Selected Base: Octal (Base 8)
3. Expected Output: "Invalid input for selected base." text appears in red.

### Test Case 4: Mathematical Bounds Limits

Input 1

Value

99999999999999999999

Base System

Decimal (Base 10)

Binary

10101101011111000111010111  
10001011010110001100001  
111111111111111111

Octal

1265707274265430377777  
7

Decimal

99999999999999999999

Hexadecimal

56BC75E2D630FFFFFF

1. Input Value: 9999999999999999999
2. Selected Base: Decimal (Base 10)
3. Expected Output: Hexadecimal correctly outputs 56BC75E2D63100000 instead of converting the string into scientific notation.

## Test Case 5: Operator Precedence (Binary + Octal + Decimal)

The screenshot shows the 'System Converter & Calculator' application. The 'Arithmetic Expression' field contains  $A + B * C$ . Below it, three input boxes are visible: 'Input A' with value 1010 (Base 2), 'Input B' with value 20 (Base 8), and 'Input C' with value 2 (Base 10). The 'Final Arithmetic Result' section displays the equation breakdown:  $(1010)_2 + (20)_8 \times (2)_{10} =$ . The results are shown in four boxes: Binary 101010, Octal 52, Decimal 42, and Hexadecimal 2A.

System Converter & Calculator

Arithmetic Expression

Use variables (A, B, C, ...) and operators (+, -, \*, /, parentheses). Example:  $(A + B - C) * D$

$A + B * C$

Input A

Value: 1010 Base System: Binary (Base 2)

Binary: 1010 Octal: 12 Decimal: 10 Hexadecimal: A

Input B

Value: 20 Base System: Octal (Base 8)

Binary: 10000 Octal: 20 Decimal: 16 Hexadecimal: 10

Input C

Value: 2 Base System: Decimal (Base 10)

Binary: 10 Octal: 2 Decimal: 2 Hexadecimal: 2

+ Add Input Number

Final Arithmetic Result

EQUATION BREAKDOWN

$(1010)_2 + (20)_8 \times (2)_{10} =$

Binary: 101010 Octal: 52

Decimal: 42 Hexadecimal: 2A

1. Inputs: Input A = 1010 (Base 2), Input B = 20 (Base 8), Input C = 2 (Base 10)
2. Arithmetic Expression:  $A + B * C$
3. Expected Equation Breakdown:  $(1010)_2 + (20)_8 \times (2)_{10}$
4. Expected Logic: Multiplication executes before addition.  $10 + (16 \times 2) = 42$ .
5. Final Arithmetic Result: Binary: 101010, Octal: 52, Decimal: 42, Hexadecimal: 2A.

## Test Case 6: Parenthetical Grouping (Binary + Decimal + Hexadecimal)

The first screenshot shows the 'System Converter & Calculator' interface. At the top, an 'Arithmetic Expression' field contains  $(A - B) / C$ . Below it are three input sections: 'Input A' with a value of 10000 in Binary (Base 2), 'Input B' with a value of 4 in Decimal (Base 10), and 'Input C' with a value of 2 in Hexadecimal (Base 16). Each input section has buttons to convert the value to other bases: Binary, Octal, Decimal, and Hexadecimal.

The second screenshot shows the 'Final Arithmetic Result' section. It displays the equation breakdown:  $((10000)_2 - (4)_{10}) \div (2)_{16} =$ . Below this, the results are shown in four boxes: Binary 110, Octal 6, Decimal 6, and Hexadecimal 6.

1. Inputs: Input A = 10000 (Base 2), Input B = 4 (Base 10), Input C = 2 (Base 16)
2. Arithmetic Expression:  $(A - B) / C$
3. Expected Equation Breakdown:  $((10000)_2 - (4)_{10}) \div (2)_{16}$
4. Expected Logic: Parentheses force subtraction before division.  $(16 - 4) \div 2 = 6$ .
5. Final Arithmetic Result: Binary: 110, Octal: 6, Decimal: 6, Hexadecimal: 6.

## Test Case 7: Mixed Operators (Octal + Decimal + Hexadecimal)

The screenshot shows the 'System Converter & Calculator' application. At the top, there's an 'Arithmetic Expression' field containing  $A * B - C$ . Below it are three input sections: 'Input A' (Value: 12, Base System: Octal (Base 8)), 'Input B' (Value: 5, Base System: Decimal (Base 10)), and 'Input C' (Value: F, Base System: Hexadecimal (Base 16)). Each input section has buttons for Binary, Octal, Decimal, and Hexadecimal. The application is running in a Chrome browser window.

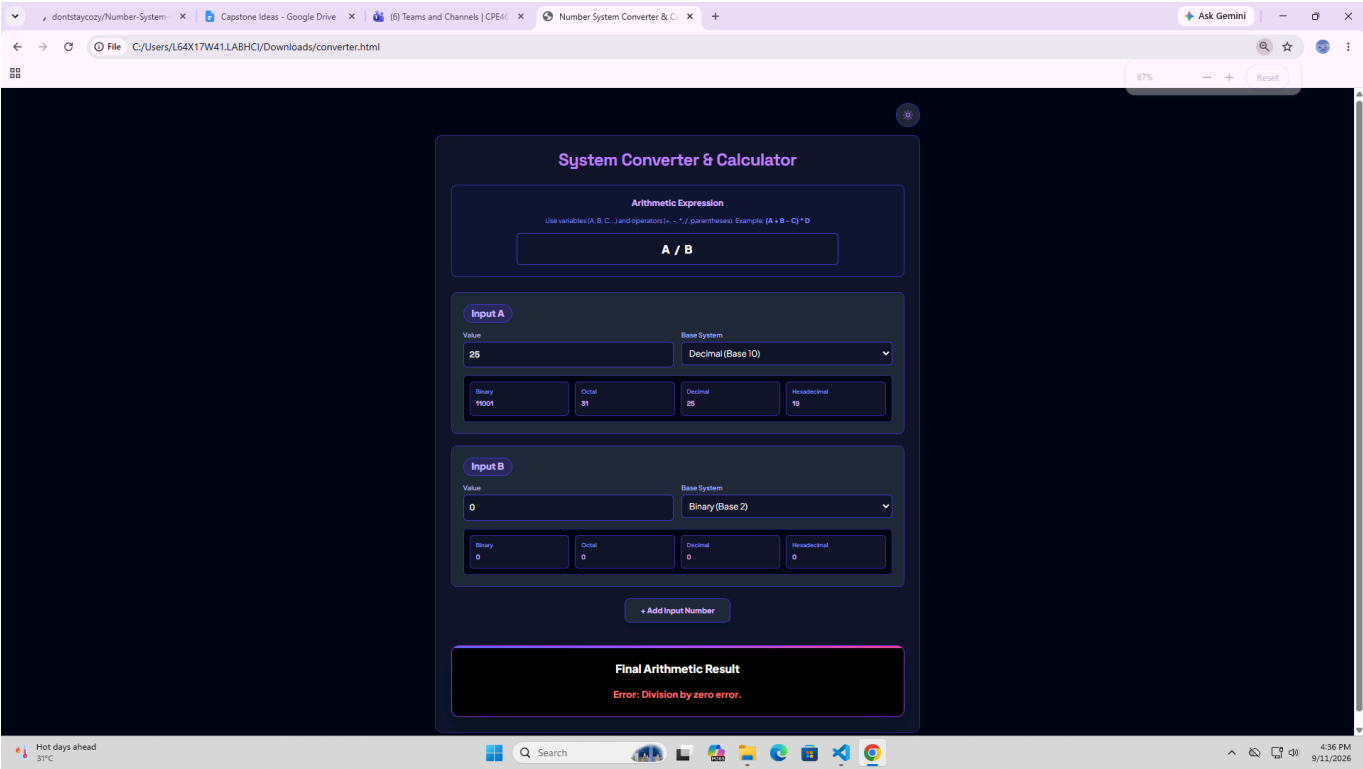
1. Inputs: **Input A** = 12 (Base 8), **Input B** = 5 (Base 10), **Input C** = F (Base 16)
2. Arithmetic Expression:  $A * B - C$
3. Expected Equation Breakdown:  $(12)_8 \times (5)_{10} - (F)_{16}$
4. Expected Logic: Multiplication executes before subtraction.  $(10 \times 5) - 15 = 35$ .
5. Final Arithmetic Result: Binary: 100011, Octal: 43, Decimal: 35, Hexadecimal: 23.

## Test Case 8: Complex Multi-Variable Expression (Binary + Octal + Hexadecimal + Binary)

The screenshot shows a web application titled "System Converter & Calculator". It features an "Arithmetic Expression" input field containing  $(A + B - C) * D$ . Below this are three input sections: "Input A" with value "1100" and base "Binary (Base 2)", "Input B" with value "10" and base "Octal (Base 8)", and "Input C" with value "A" and base "Hexadecimal (Base 16)". A fourth input section, "Input D", is partially visible with value "11" and base "Binary (Base 2)". At the bottom, a "Final Arithmetic Result" section displays the equation breakdown:  $((1100)_2 + (10)_8 - (A)_{16}) \times (11)_2 =$ . The results are shown in four boxes: Binary "11110", Octal "36", Decimal "30", and Hexadecimal "1E".

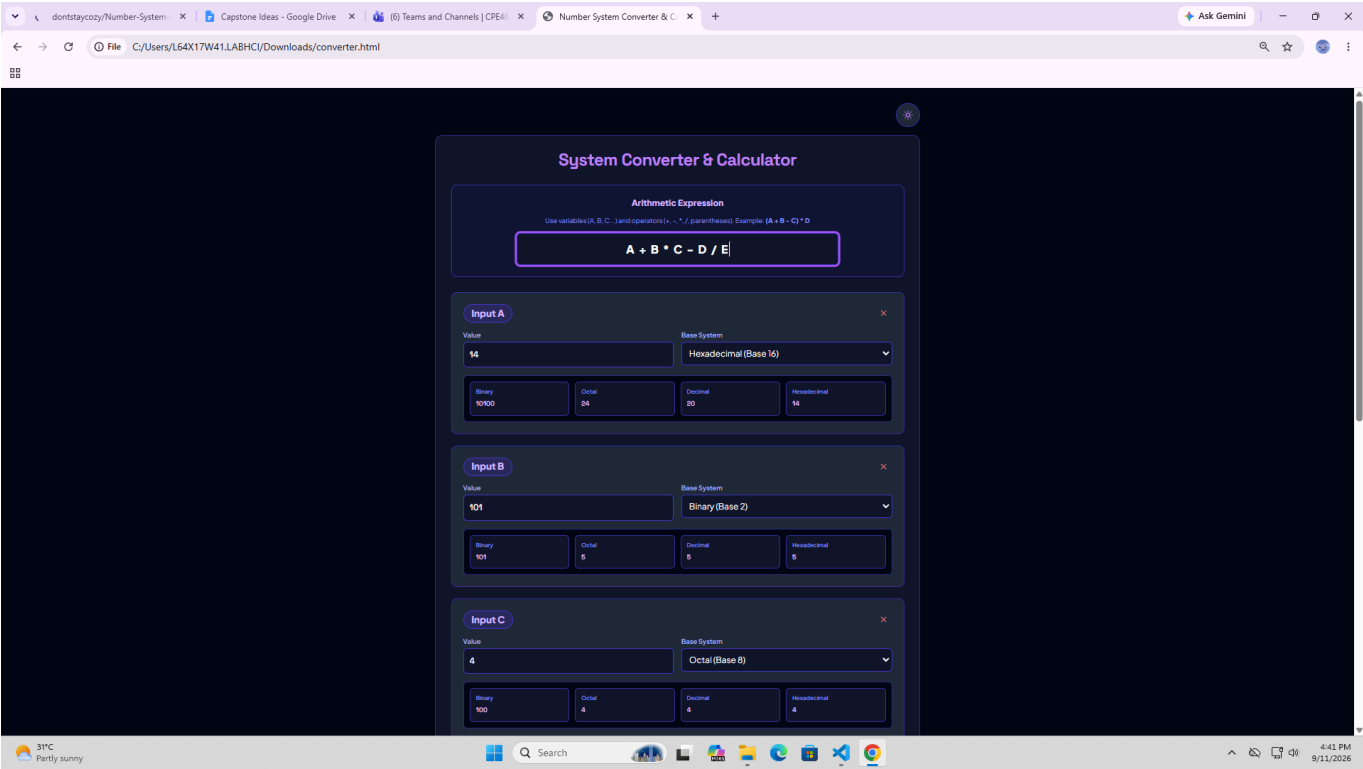
1. Inputs: Input A = 1100 (Base 2), Input B = 10 (Base 8), Input C = A (Base 16), Input D = 11 (Base 2)
2. Arithmetic Expression:  $(A + B - C) * D$
3. Expected Equation Breakdown:  $((1100)_2 + (10)_8 - (A)_{16}) \times (11)_2$
4. Expected Logic: Left-to-right inside parentheses, followed by multiplication.  $(12 + 8 - 10) \times 3 = 30$ .
5. Final Arithmetic Result: Binary: 11110, Octal: 36, Decimal: 30, Hexadecimal: 1E.

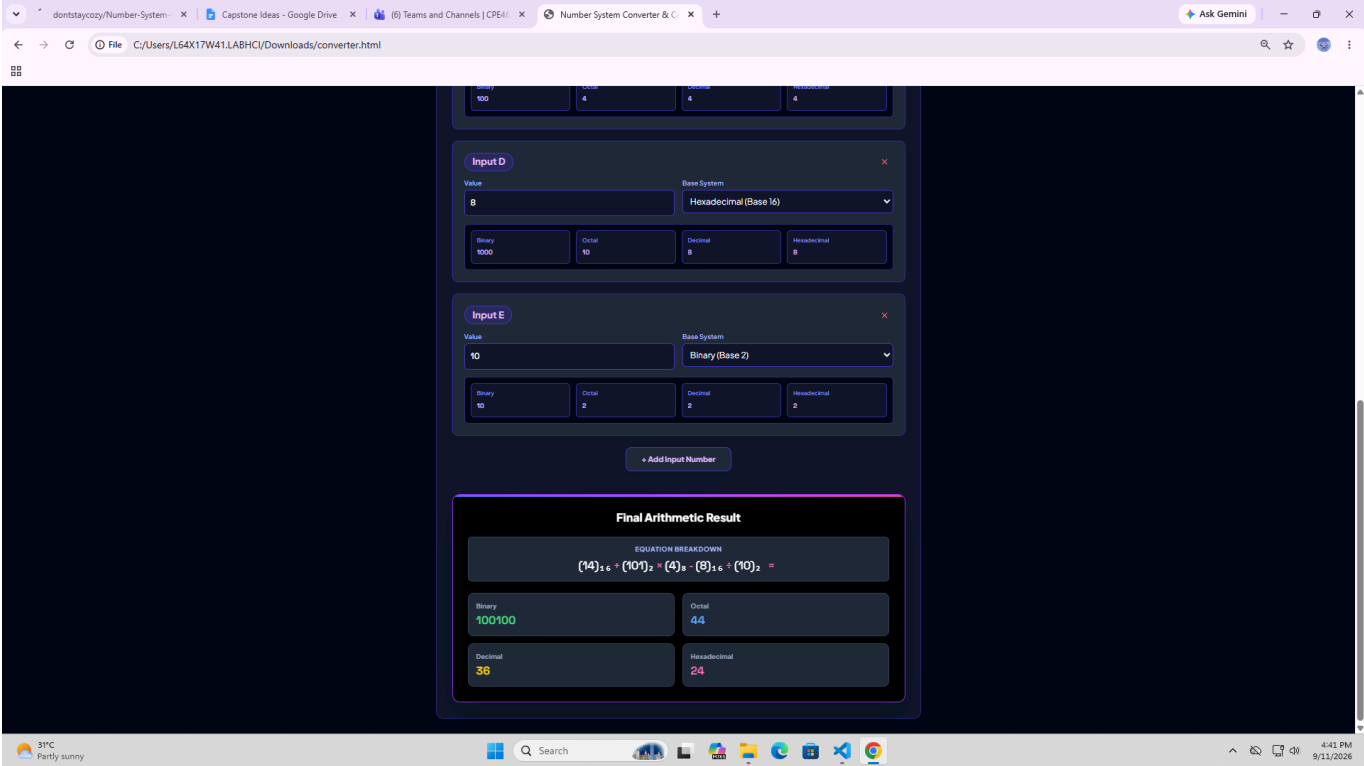
Test Case 9: Arithmetic Error Handling (Division by Zero)



- 1. Inputs: **Input A** = 25 (Base 10), **Input B** = 0 (Base 2)
- 2. Arithmetic Expression: **A / B**
- 3. Expected Equation Breakdown: None.
- 4. Expected Output: Red error message stating "Error: Division by zero error." prevents the calculation.

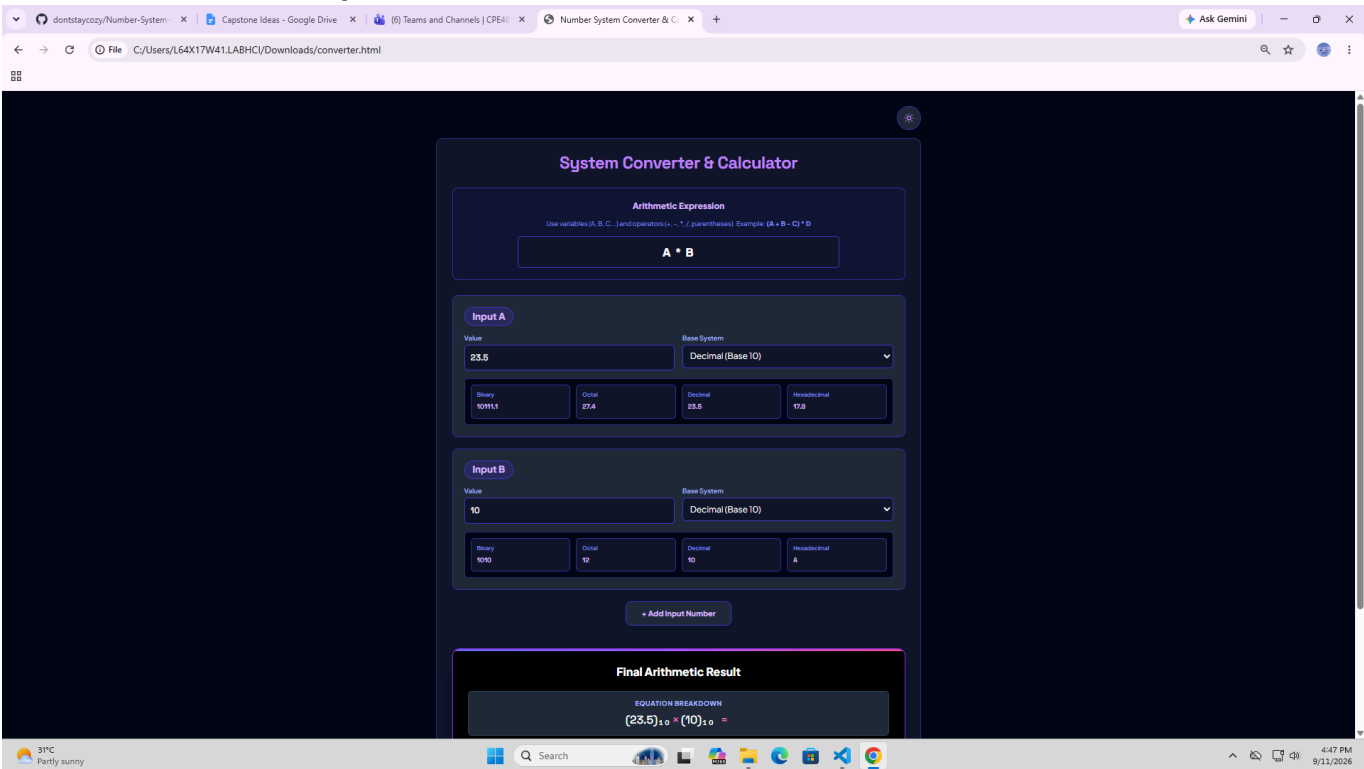
Test Case 10: Comprehensive Order of Operations (Mixed Bases)

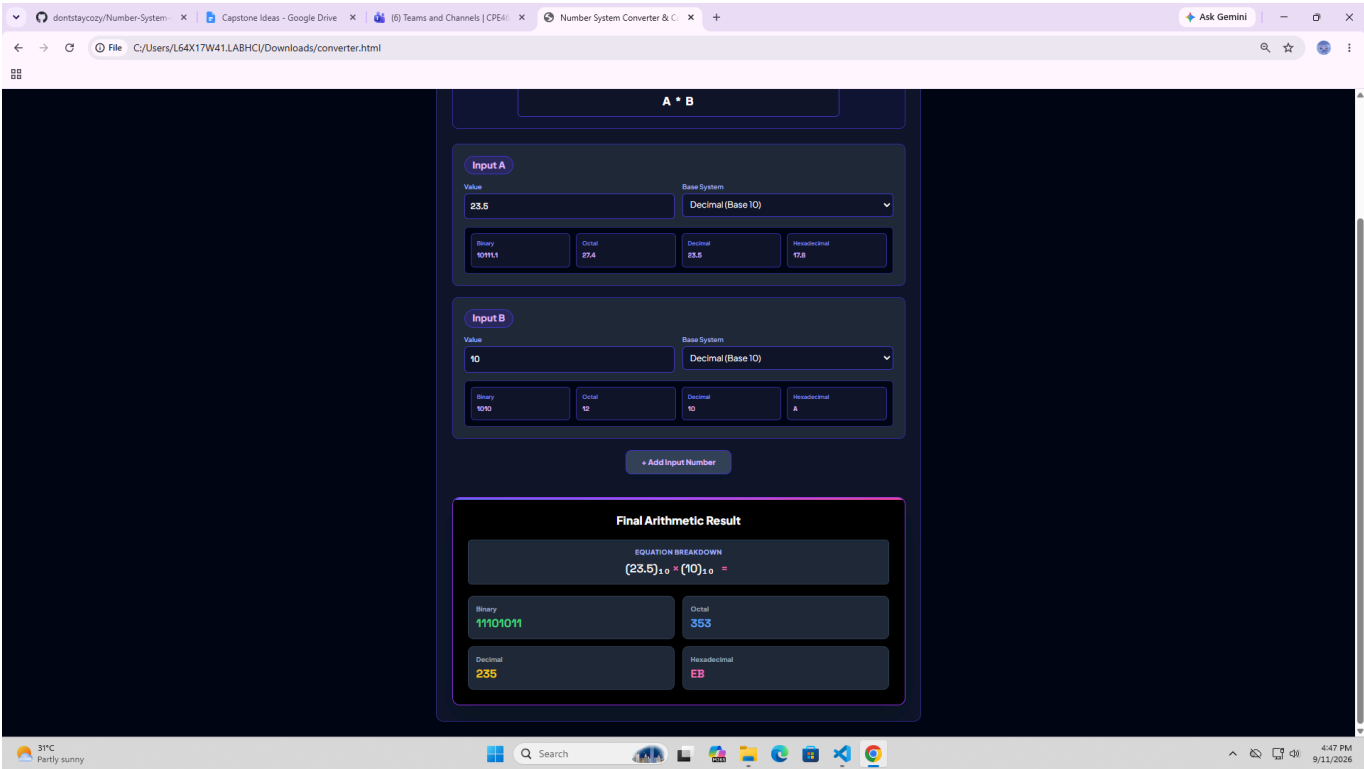




- 1. Inputs: **Input A** = 14 (Hexadecimal), **Input B** = 101 (Binary), **Input C** = 4 (Octal), **Input D** = 8 (Hexadecimal), **Input E** = 10 (Binary)
- 2. Arithmetic Expression:  $A + B * C - D / E$
- 3. Expected Equation Breakdown:  $(14)_{16} + (101)_2 \times (4)_8 - (8)_{16} \div (10)_2$
- 4. Expected Logic: Base conversions equate to  $(20 + 5 \times 4 - 8 \div 2)$ . Multiplication and division execute first  $(5 \times 4 = 20)$  and  $(8 \div 2 = 4)$ , followed by addition and subtraction  $(20 + 20 - 4) = 36$ .
- 5. Final Arithmetic Result: Binary: 100100, Octal: 44, Decimal: 36, Hexadecimal: 24.

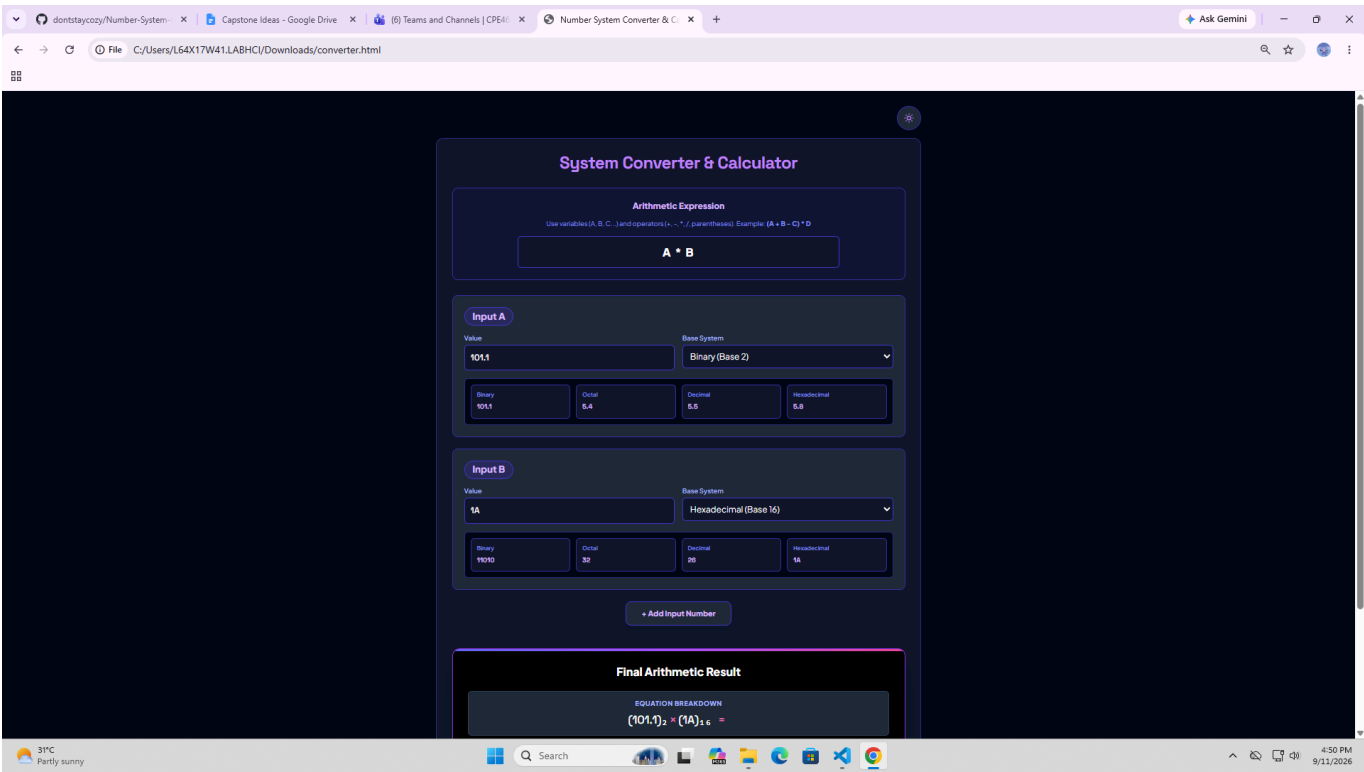
Test Case 11: Fractional Input Calculation

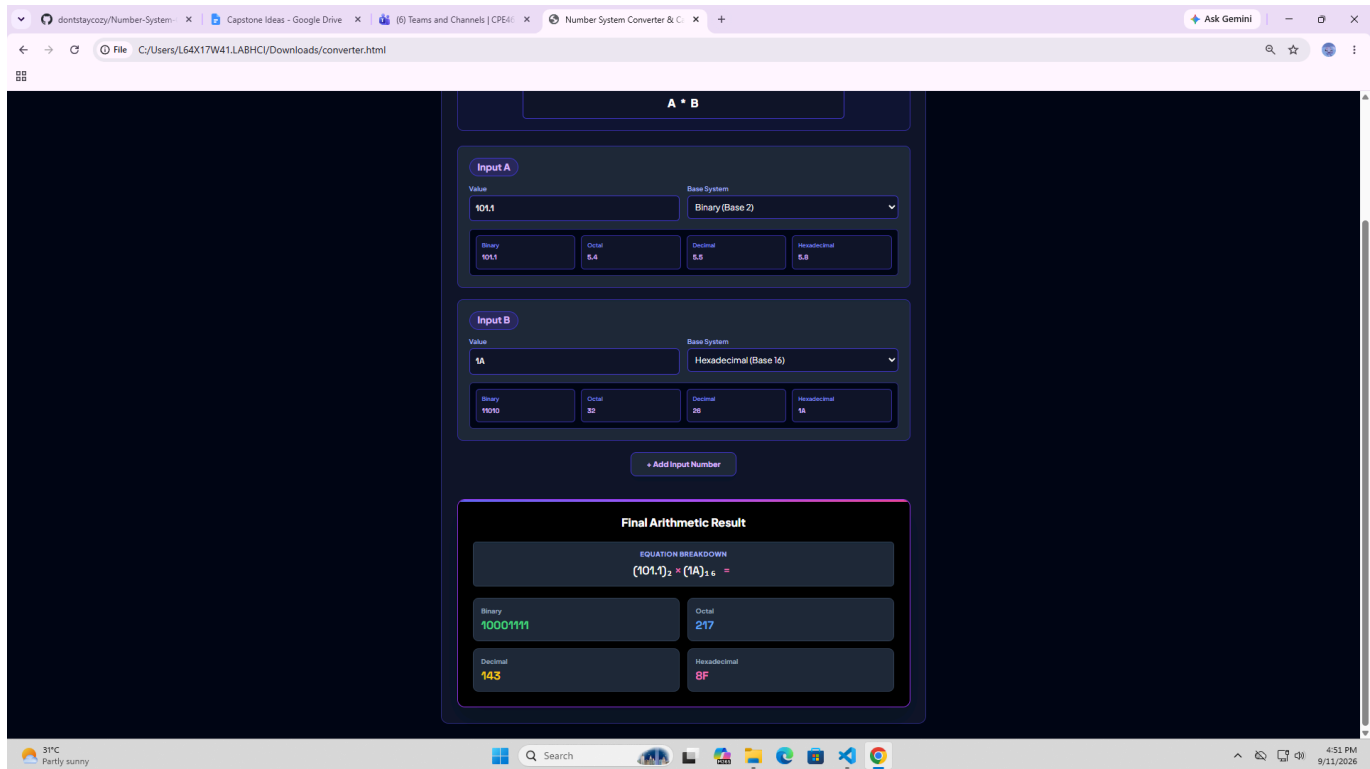




- 1. Inputs: Input A = 23.5 (Decimal), Input B = 10 (Decimal)
- 2. Arithmetic Expression: A \* B
- 3. Expected Equation Breakdown:  $(23.5)_{10} \times (10)_{10}$
- 4. Expected Logic: Multiplication of a fractional decimal by a whole decimal.  $23.5 \times 10 = 235$ .
- 5. Final Arithmetic Result: Binary: 11101011, Octal: 353, Decimal: 235, Hexadecimal: EB.

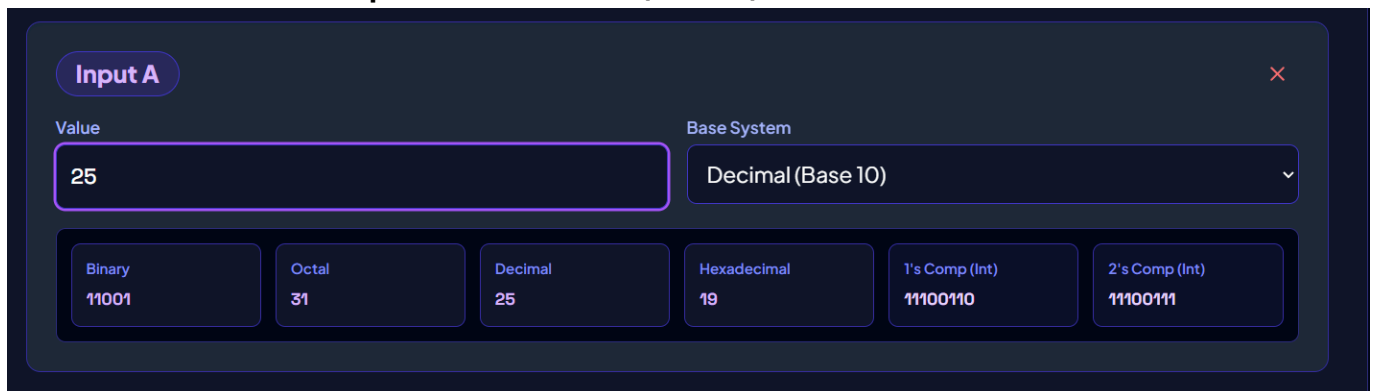
Test Case 12: Non-Decimal Fractional Input Calculation





1. Inputs: **Input A** = 101.1 (Binary), **Input B** = 1A (Hexadecimal)
2. Arithmetic Expression:  $A + B$
3. Expected Equation Breakdown:  $(101.1)_2 + (1A)_{16}$
4. Expected Logic: Base conversions equate to  $(5.5 + 26)$ . The fractional binary successfully adds to the whole hexadecimal.  $5.5 + 26 = 31.5$ .
5. Final Arithmetic Result: Binary: 11111.1, Octal: 37.4, Decimal: 31.5, Hexadecimal: 1F.8.

### Test Case 13: Standard Complement Generation (Base 10)



1. Inputs: **Input A** = 25 (Decimal)
2. Expected Logic: The system computes standard base conversions and dynamically generates the 8-bit padded 1's and 2's complements for the positive integer.
3. Expected Output: Binary: 11001, Octal: 31, Decimal: 25, Hexadecimal: 19, 1's Comp (Int): 11100110, 2's Comp (Int): 11100111.

## Test Case 14: Complement Subtraction (Positive Result / Minuend > Subtrahend)

The screenshot displays the 'Number System Converter & Calculator' application. The 'Input A' section shows a value of 15 in the decimal system, with its binary (1111), octal (17), hexadecimal (F), 1's complement (11110000), and 2's complement (11110001) representations. The 'Input B' section shows a value of 5 in the decimal system, with its binary (101), octal (5), hexadecimal (5), 1's complement (1111010), and 2's complement (1111011) representations. Below these inputs, the 'Complement Subtraction Method' is detailed. It shows the step-by-step process for subtracting 5 from 15 using both 1's and 2's complement methods. For the 1's complement method, the 1's complement of 5 (1111010) is added to 15 (00001111), resulting in a carry of 1 and a final result of 00001010 (10). For the 2's complement method, the 2's complement of 5 (1111011) is added to 15 (00001111), resulting in a carry of 1 (discarded) and a final result of 00001010 (10).

**Input A**

Value: 15 Base System: Decimal (Base 10)

Binary: 1111 Octal: 17 Decimal: 15 Hexadecimal: F 1's Comp (Int): 11110000 2's Comp (Int): 11110001

**Input B**

Value: 5 Base System: Decimal (Base 10)

Binary: 101 Octal: 5 Decimal: 5 Hexadecimal: 5 1's Comp (Int): 1111010 2's Comp (Int): 1111011

**Complement Subtraction Method**

Select two variables to demonstrate step-by-step subtraction (Minuend - Subtrahend) using complements. Uses absolute integer parts.

Var A - Var B

**1's Complement Method**

M: 00001111  
S: 00000101  
1's Comp(S): 1111010

Add M + 1's Comp(S):

```
00001111
+1111010
-----
1 00001001
```

Carry = 1. Add End-Around Carry (+1).

Final: 00001010

**2's Complement Method**

M: 00001111  
S: 00000101  
2's Comp(S): 1111011

Add M + 2's Comp(S):

```
00001111
+1111011
-----
1 00001010
```

Discard carry out.

Final: 00001010

1. Inputs: Set Minuend dropdown to Variable A (Decimal 15), Set Subtrahend to Variable B (Decimal 5).
2. Expected Logic: The tool processes 15 - 5 using 8-bit representation.
3. 1's Complement Output: Converts S(5) to 00000101, 1's comp 1111010. Adds to M(15) 00001111. Yields carry=1. Adds end-around carry to yield 00001010 (10).
4. 2's Complement Output: Converts S(5) to 2's comp 1111011. Adds to M(15). Yields carry=1 (discarded), resulting directly in 00001010 (10).

## Test Case 15: Complement Subtraction (Negative Result / Minuend < Subtrahend)

The screenshot shows a web application titled "Number System Converter & Calculator". It has two main input sections, "Input A" and "Input B".

**Input A:**

- Value: 5
- Base System: Decimal (Base 10)
- Binary: 101
- Octal: 5
- Decimal: 5
- Hexadecimal: 5
- 1's Comp (Int): 1111010
- 2's Comp (Int): 1111011

**Input B:**

- Value: 15
- Base System: Decimal (Base 10)
- Binary: 1111
- Octal: 17
- Decimal: 15
- Hexadecimal: F
- 1's Comp (Int): 11110000
- 2's Comp (Int): 11110001

At the bottom, there is a button labeled "+ Add Input Number". The Windows taskbar at the bottom shows the time as 11:29 AM on 9/19/2026.

1. Inputs: Set Minuend dropdown to Variable A (Decimal 5), Set Subtrahend to Variable B (Decimal 15).
2. Expected Logic: The tool processes  $5 - 15$  using 8-bit representation.
3. 1's Complement Output: Converts S(15) to 00001111, 1's comp 11110000. Adds to M(5) 00000101. Yields no carry. Final result is a negative number in 1's complement form: 11110101.
4. 2's Complement Output: Converts S(15) to 2's comp 11110001. Adds to M(5). Yields no carry. Final result is a negative number in 2's complement form: 11110110.

## Test Case 16: Final Arithmetic Result with Complements

The screenshot shows the same web application as before, but with different input values.

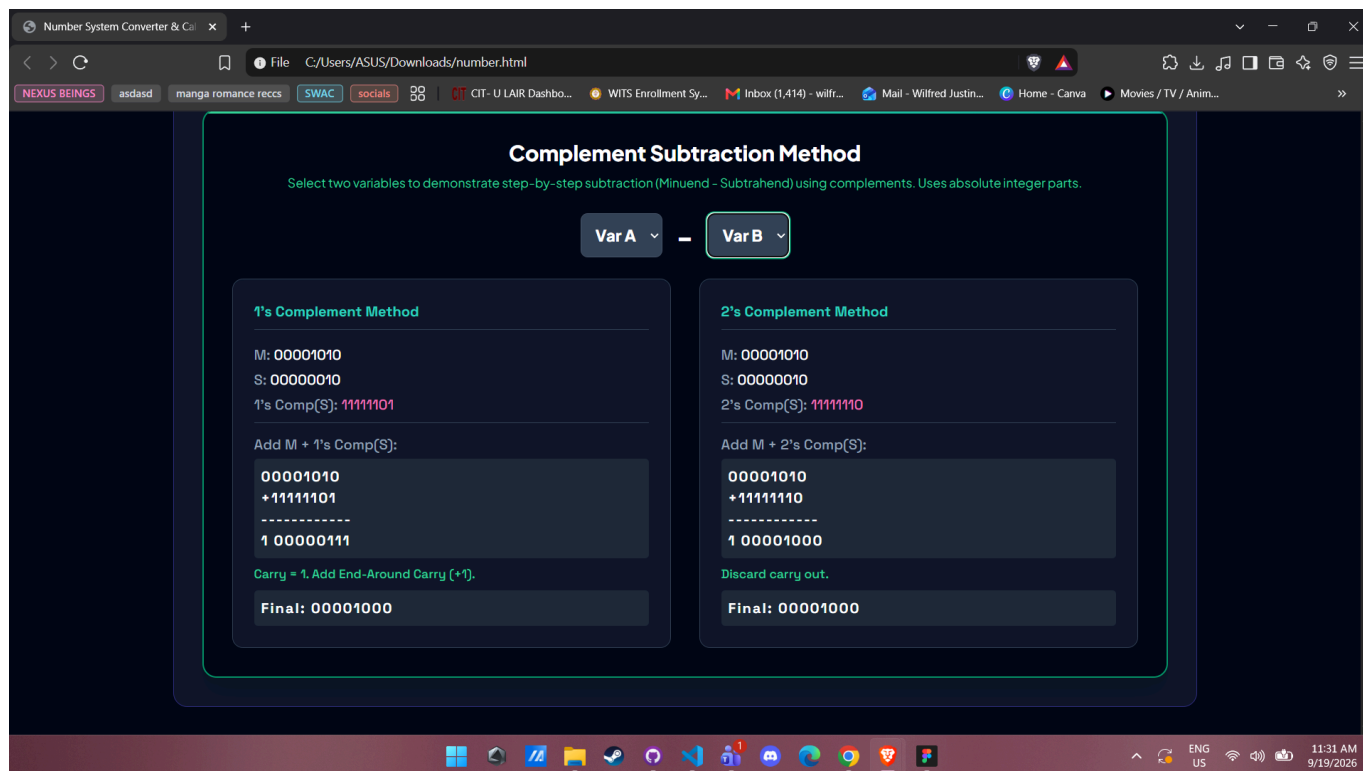
**Input A:**

- Value: 10
- Base System: Decimal (Base 10)
- Binary: 1010
- Octal: 12
- Decimal: 10
- Hexadecimal: A
- 1's Comp (Int): 11110101
- 2's Comp (Int): 11110110

**Input B:**

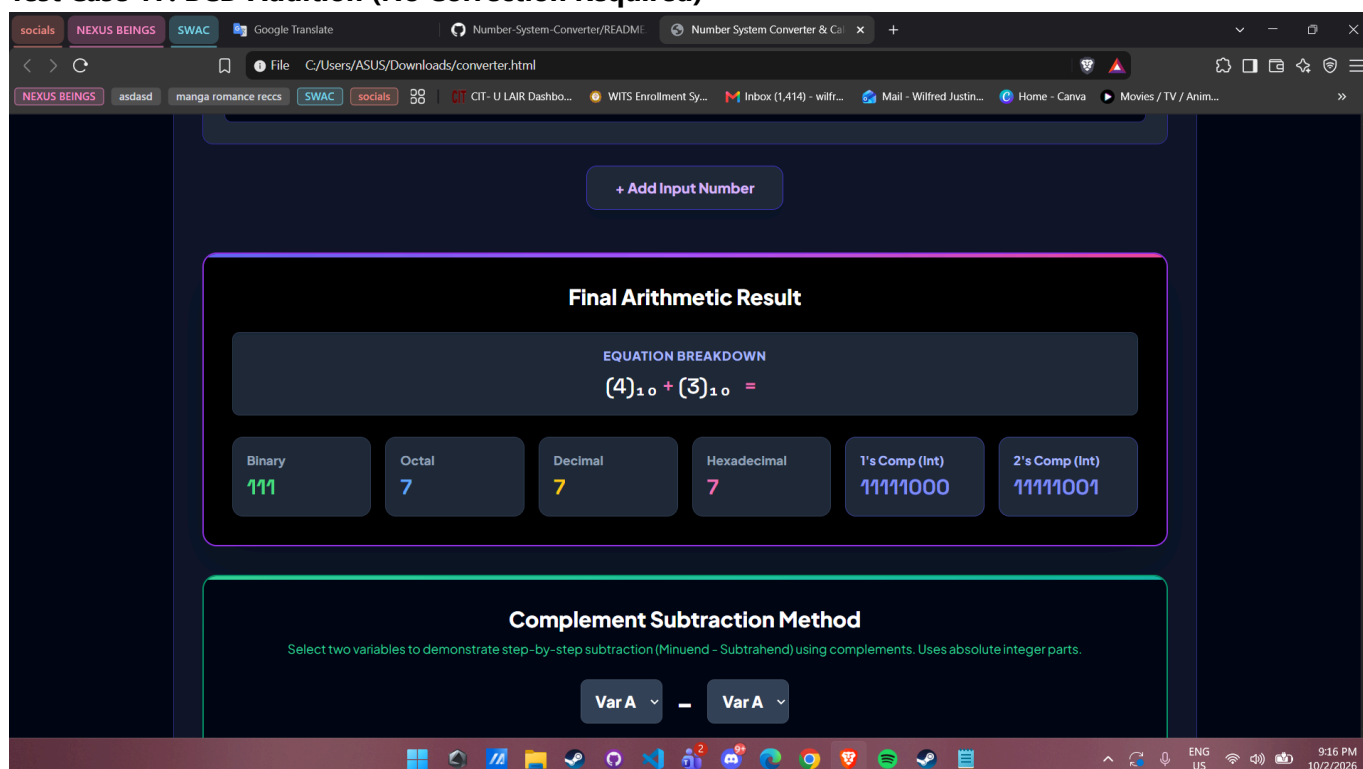
- Value: 2
- Base System: Decimal (Base 10)
- Binary: 10
- Octal: 2
- Decimal: 2
- Hexadecimal: 2
- 1's Comp (Int): 11111101
- 2's Comp (Int): 11111110

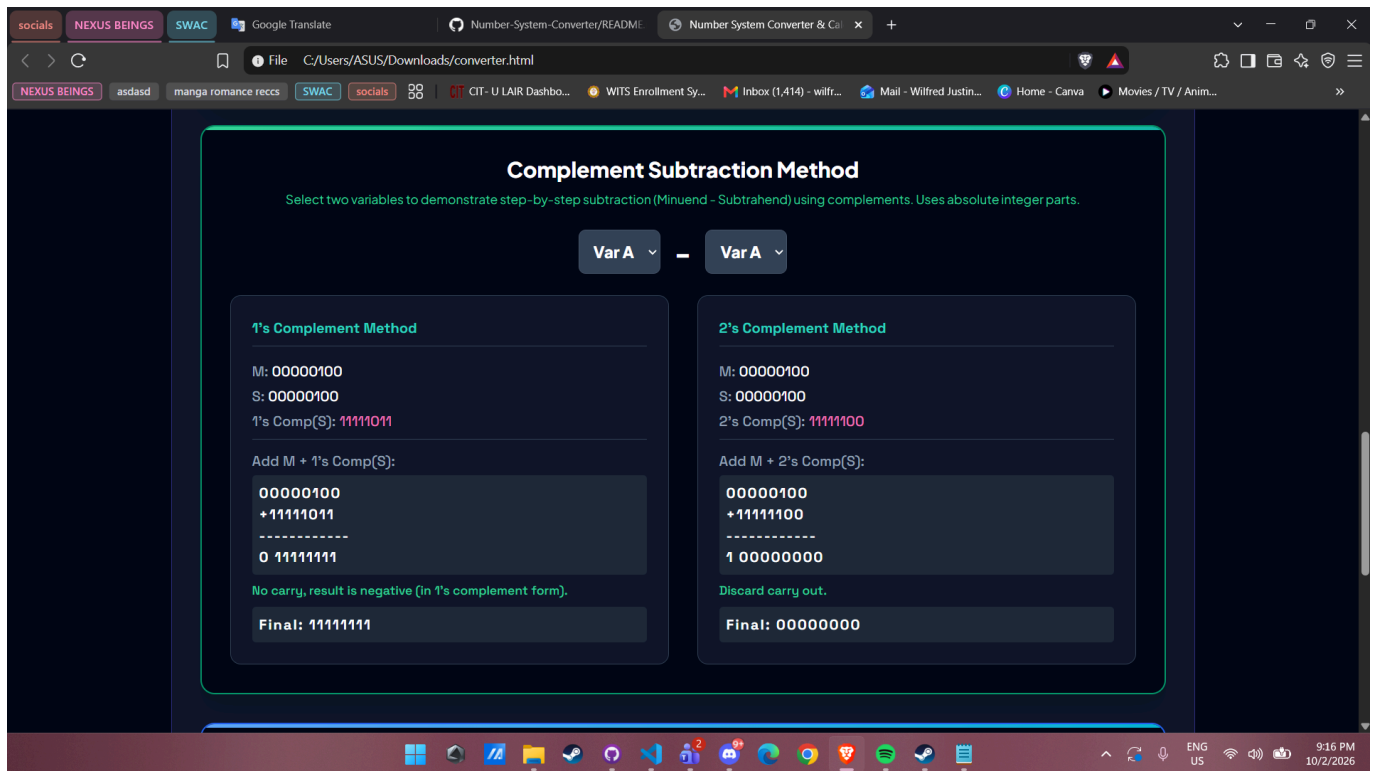
At the bottom, there is a button labeled "+ Add Input Number". The Windows taskbar at the bottom shows the time as 11:31 AM on 9/19/2026.



1. Inputs: Input A = 10 (Decimal), Input B = 2 (Decimal)
2. Arithmetic Expression:  $A - B$
3. Expected Equation Breakdown:  $(10)_{10} - (2)_{10}$
4. Expected Logic: The system computes  $10 - 2 = 8$ , then dynamically generates the complements for the final computed result.
5. Final Arithmetic Result: Binary: 1000, Octal: 10, Decimal: 8, Hexadecimal: 8, 1's Comp (Int): 11110111, 2's Comp (Int): 11111000.

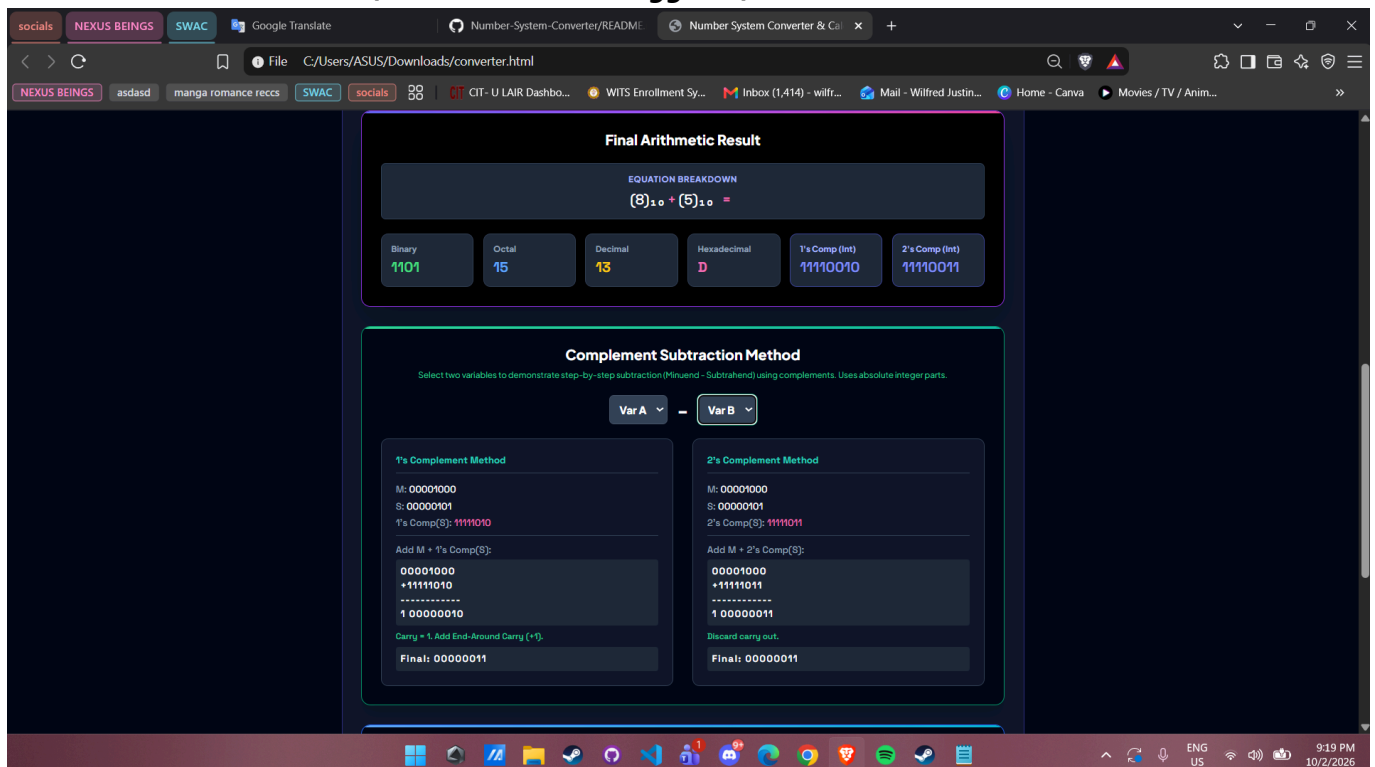
### Test Case 17: BCD Addition (No Correction Required)

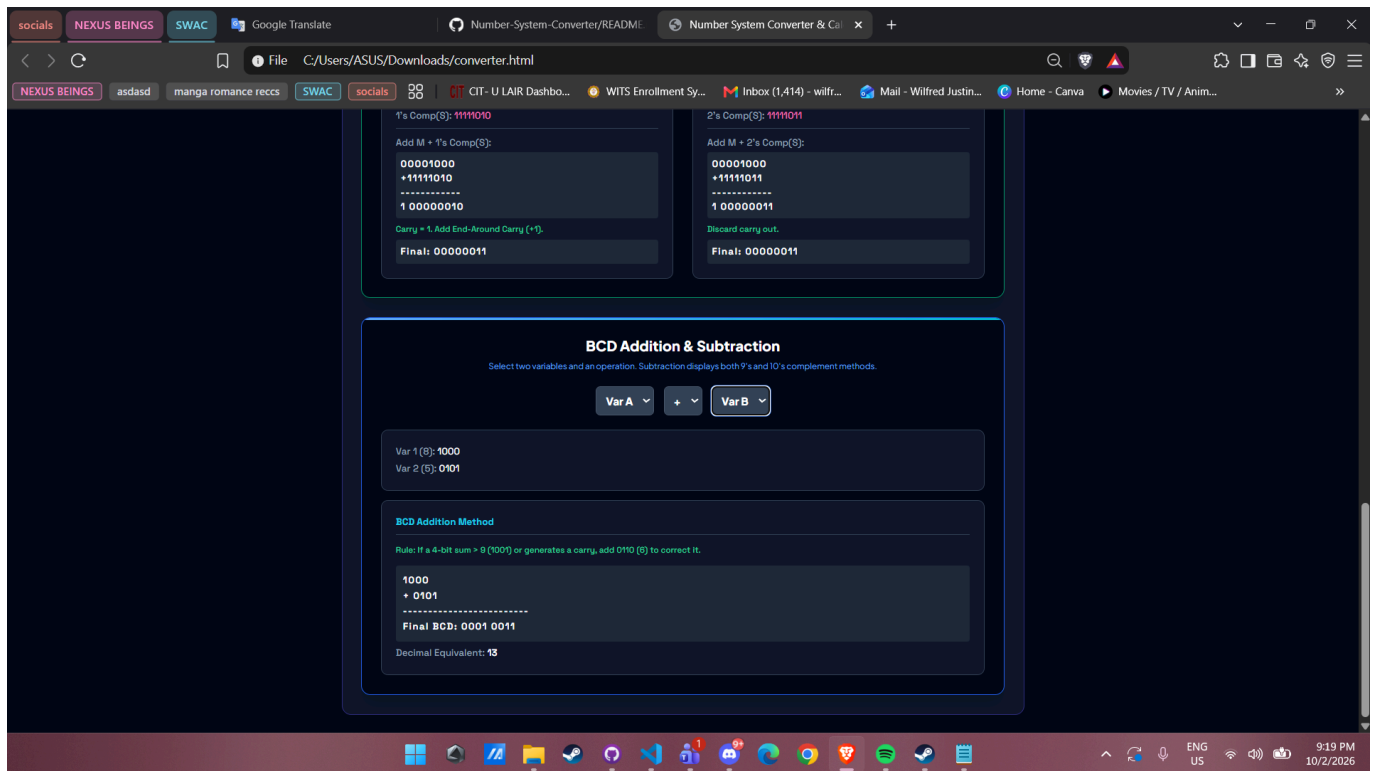




1. Inputs: Set BCD Var 1 to **A** (Decimal **4**), Operator to **+**, BCD Var 2 to **B** (Decimal **3**).
2. Expected Logic: The system computes **0100 + 0011**. Since the sum is **0111** (7) and is less than 9, no correction factor is applied.
3. Expected Output: Final BCD is **0111**. Decimal Equivalent is **7**.

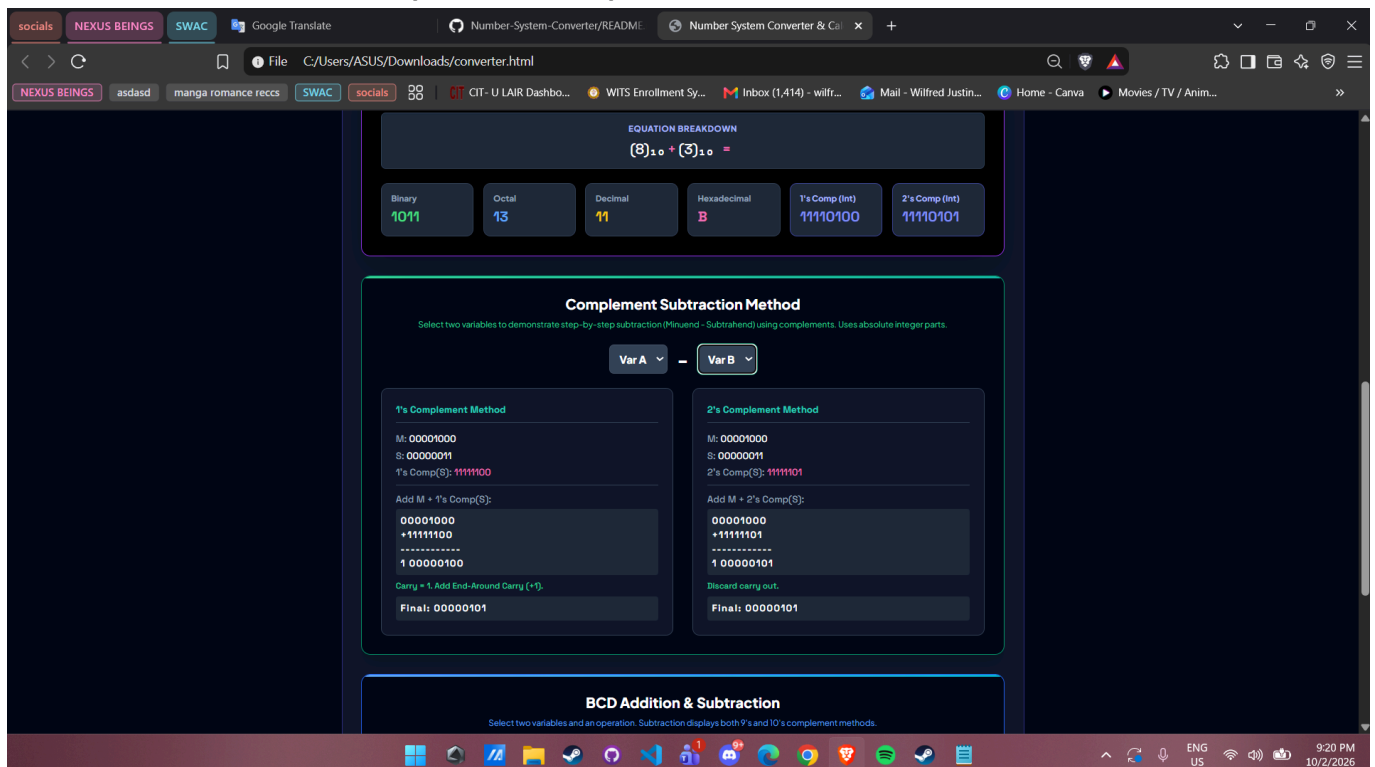
### Test Case 18: BCD Addition (Correction Factor Triggered)

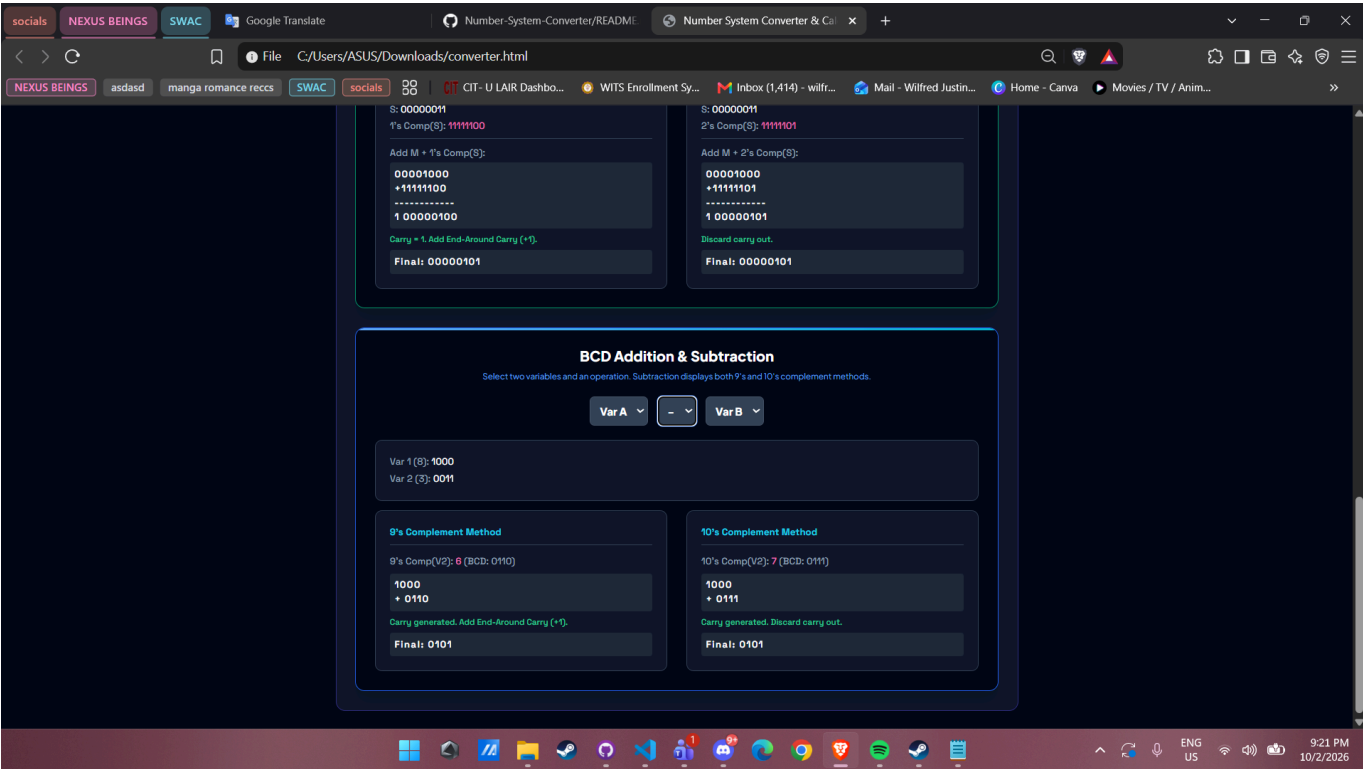




1. Inputs: Set BCD Var 1 to **A** (Decimal 8), Operator to **+**, BCD Var 2 to **B** (Decimal 5).
2. Expected Logic: The system computes **1000 + 0101**, yielding **1101** (13). Since  $13 > 9$ , the **+0110** (6) correction factor is applied.
3. Expected Output: Final BCD is **0001 0011**. Decimal Equivalent is **13**.

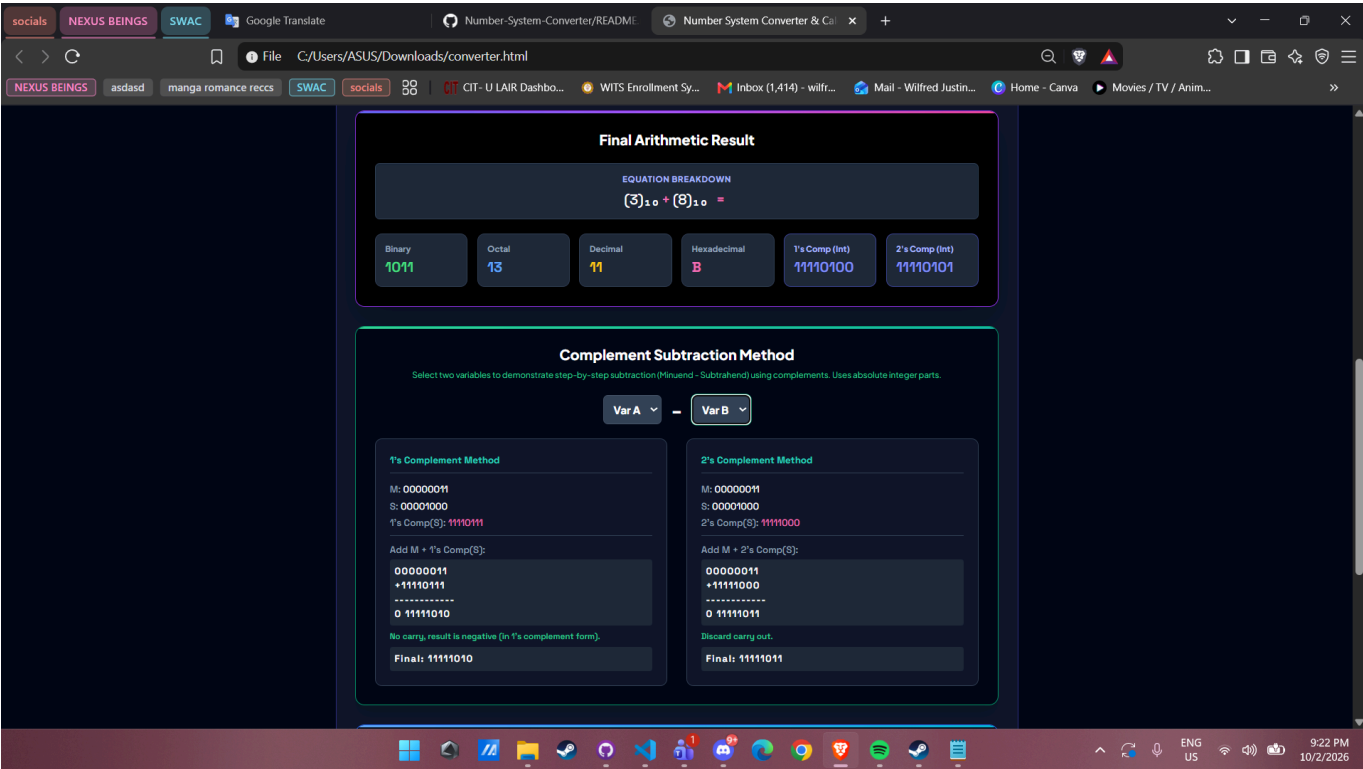
### Test Case 19: BCD Subtraction (Positive Result)

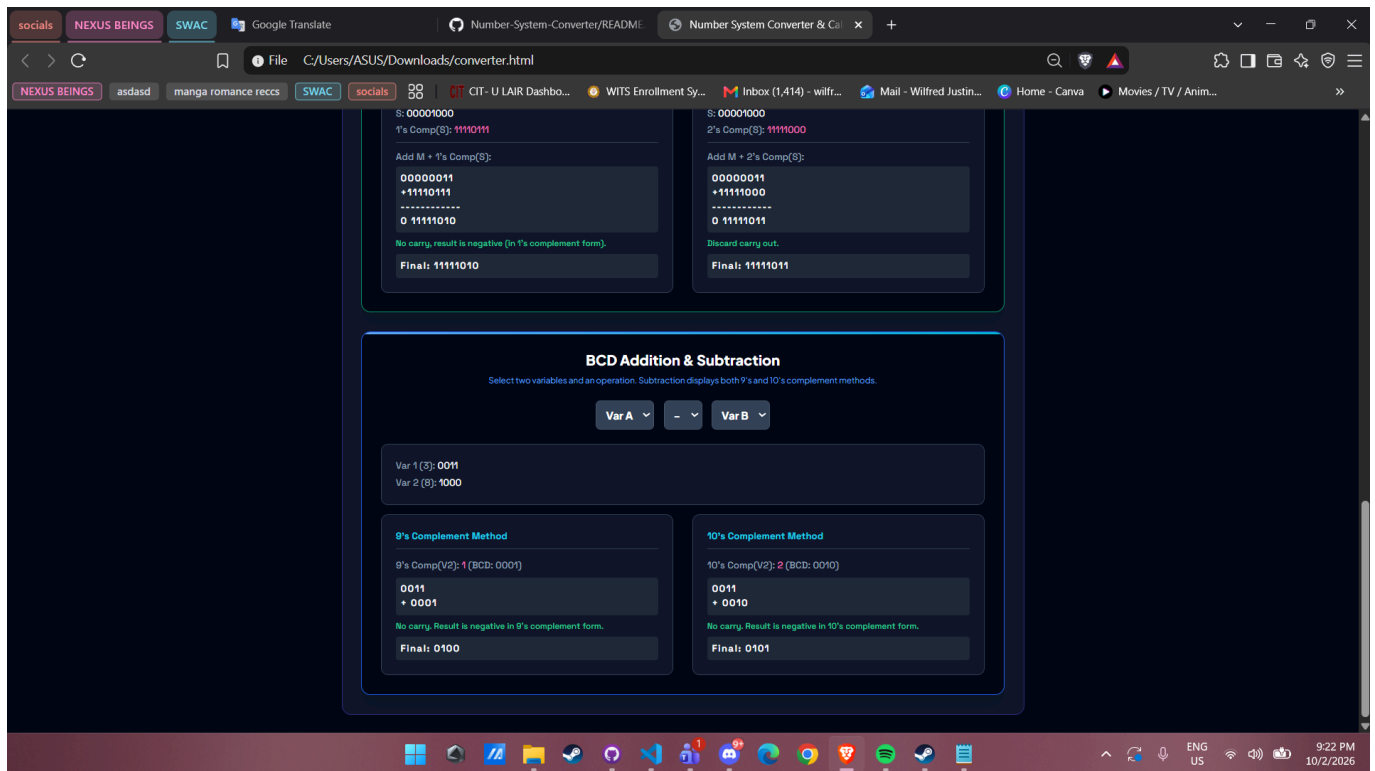




- 1. Inputs: Set BCD Var 1 to **A** (Decimal 8), Operator to **-**, BCD Var 2 to **B** (Decimal 3).
- 2. Expected Logic: The tool converts **3** to its 9's and 10's complements.
- 3. 9's Complement Output: Generates a carry. Adds End-Around Carry (+1) to yield **0101** (5).
- 4. 10's Complement Output: Generates a carry. Discards carry out to yield **0101** (5).

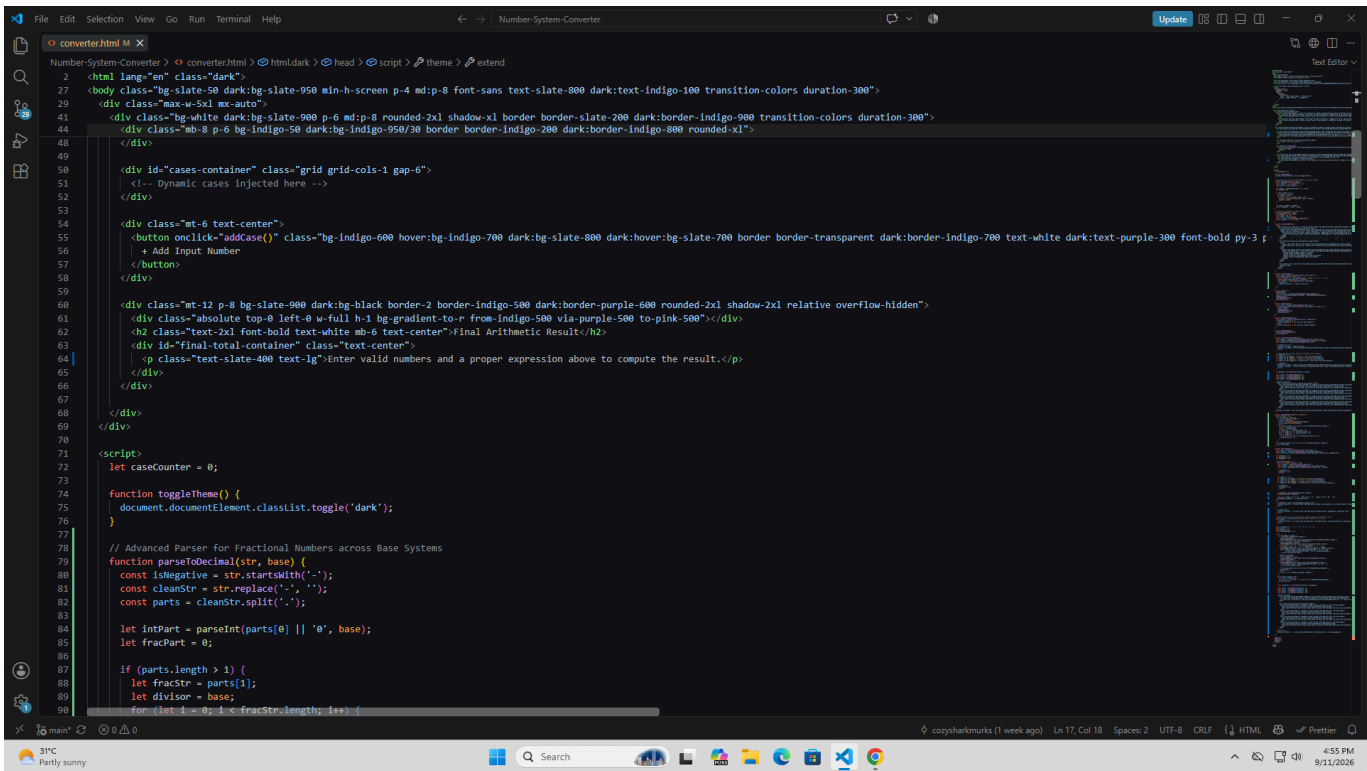
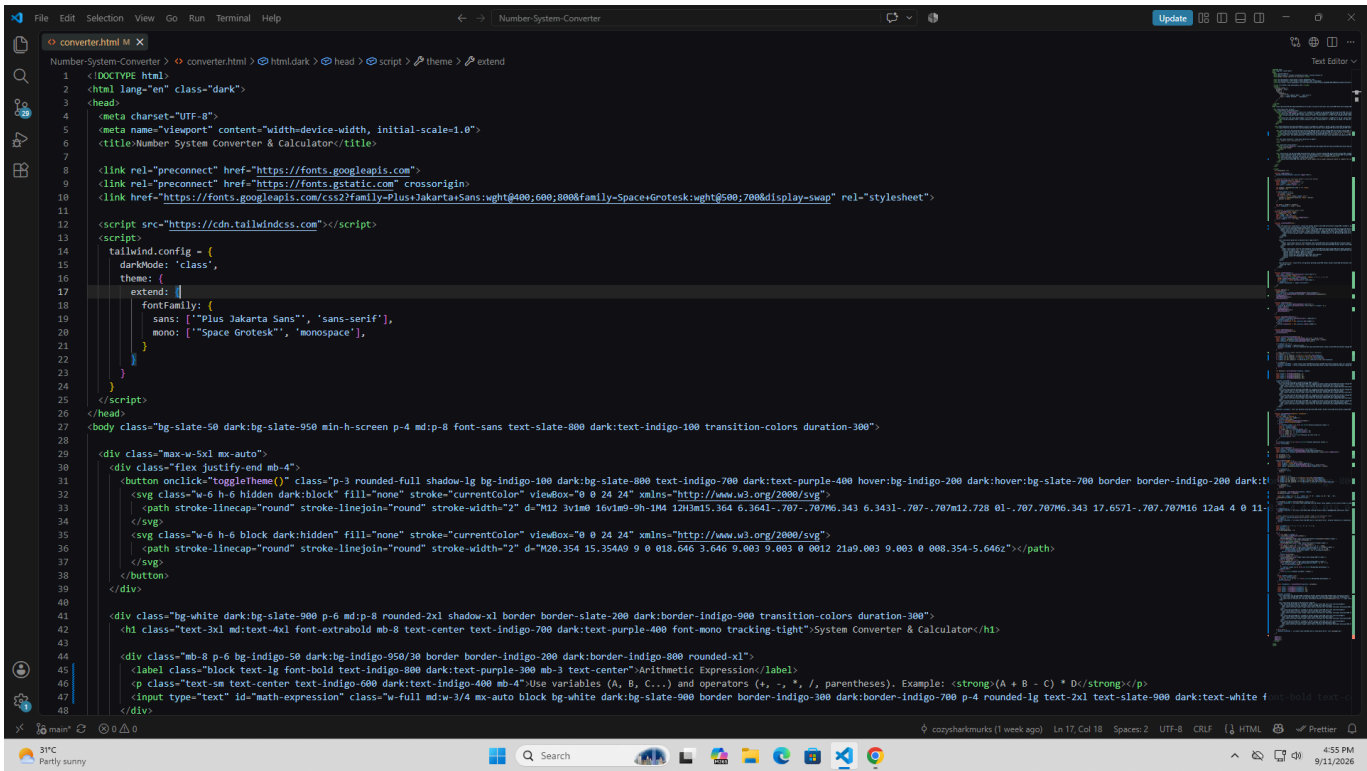
Test Case 20: BCD Subtraction (Negative Result)





1. Inputs: Set BCD Var 1 to **A** (Decimal **3**), Operator to **-**, BCD Var 2 to **B** (Decimal **8**).
2. Expected Logic: Subtracting a larger number yields no carry.
3. Expected Output: The system identifies the lack of a carry and leaves the final result in its negative 9's complement and 10's complement forms without attempting to apply an End-Around Carry.

## Code Screenshots



```
File Edit Selection View Go Run Terminal Help
Number-System-Converter > converter.html > html:dark > head > script > theme > extend
2 <html lang="en" class="dark">
27 <body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
71 <script>
72   function parseToDecimal(str, base) {
87     if (parts.length > 1) {
88       let fracStr = parts[1];
89       let divisor = base;
90       for (let i = 0; i < fracStr.length; i++) {
91         fracPart += parseInt(fracStr[i], base) / divisor;
92         divisor *= base;
93       }
94     }
95
96     let total = intPart + fracPart;
97     return isNegative ? -total : total;
98   }
99
100   // Formatter to standardize output format
101   function formatBase(num, base) {
102     if (isNaN(num)) return "NaN";
103     const isNeg = num < 0;
104     const absVal = Math.abs(num);
105     let str = absVal.toString(base).toUpperCase();
106     return (isNeg ? '-' : '') + str;
107   }
108
109   function createCaseHTML(id) {
110     return `
111     <div id="case-${id}" data-var="" class="p-6 border border-slate-200 dark:border-indigo-800 rounded-xl bg-slate-50 dark:bg-slate-800 relative shadow-sm dark:shadow-inner transition-colors duration-300">
112       <div class="flex justify-between items-center mb-4">
113         <label class="case-label font-bold text-xl text-indigo-800 dark:text-purple-300 bg-indigo-100 dark:bg-indigo-900/50 px-4 py-1 rounded-full border border-indigo-200 dark:border-indigo-700">Input</label>
114         <button onclick="removeCase(${id})" class="remove-btn p-2 text-red-500 dark:text-red-400 hover:text-red-700 dark:hover:text-red-300 hover:bg-red-100 dark:hover:bg-red-900/30 rounded-lg hidden transition-colors">
115           <svg class="w-5 h-5" fill="none" stroke="currentColor" viewBox="0 0 24 24" xmlns="http://www.w3.org/2000/svg">
116             <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M6 18L18 6" />
117           </svg>
118         </button>
119       </div>
120
121       <div class="grid grid-cols-1 md:grid-cols-2 gap-4 mb-4">
122         <div>
123           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Value</label>
124           <input type="text" id="val-${id}" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg font-mono text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
125         </div>
126         <div>
127           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Base System</label>
128           <select id="base-${id}" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
129           <option value="2">Binary (Base 2)</option>
130         </div>
131       </div>
132     </div>`;
133   }
134
135   function reindexLabels() {
136     const cases = document.querySelectorAll("[id^='case-']");
137     cases.forEach((caseEl, index) => {
138       const varLetter = String.fromCharCode(65 + index); // 0 = A, 1 = B, etc.
139       caseEl.setAttribute('data-var', varLetter);
140       const labelEl = caseEl.querySelector('.case-label');
141       if (labelEl) {
142         labelEl.textContent = `Input ${varLetter}`;
143       }
144     });
145   }
146
147   function addCase() {
148     caseCounter++;
149     const container = document.getElementById('cases-container');
150     container.insertAdjacentHTML('beforeend', createCaseHTML(caseCounter));
151     reindexLabels();
152     updateRemoveButtons();
153     calculateTotal();
154   }
155
156   function removeCase(id) {
157     const caseEl = document.getElementById(`case-${id}`);
158     if (caseEl && document.querySelectorAll("[id^='case-']").length > 2) {
159       caseEl.remove();
160       reindexLabels();
161     }
162   }
163
164   // Initialize
165   reindexLabels();
166   calculateTotal();
167   updateRemoveButtons();
168   addCase();
169
170   // Event listeners
171   document.getElementById('add-btn').addEventListener('click', addCase);
172   document.getElementById('remove-btn').addEventListener('click', removeCase);
173   document.getElementById('base-select').addEventListener('change', calculateTotal);
174   document.getElementById('val-input').addEventListener('input', calculateTotal);
175
176   // Calculate Total
177   function calculateTotal() {
178     const valInput = document.getElementById('val-input');
179     const baseSelect = document.getElementById('base-select');
180     const base = parseInt(baseSelect.value, 10);
181     const val = valInput.value;
182     const decimal = parseToDecimal(val, base);
183     const formatted = formatBase(decimal, base);
184     document.getElementById('result').textContent = formatted;
185   }
186
187   // Update Remove Buttons
188   function updateRemoveButtons() {
189     const cases = document.querySelectorAll('.case-label');
190     cases.forEach((caseLabel, index) => {
191       const btnId = `remove-${index + 1}`;
192       const btn = document.getElementById(btnId);
193       if (btn) {
194         btn.classList.toggle('hidden', cases.length <= 2);
195       }
196     });
197   }
198
199   // Initial Setup
200   reindexLabels();
201   calculateTotal();
202   updateRemoveButtons();
203   addCase();
204
205   // Close Modal
206   document.getElementById('close-modal').addEventListener('click', closeModal);
207
208   // Show Modal
209   document.getElementById('show-modal').addEventListener('click', showModal);
210
211   // Modal Content
212   function showModal() {
213     const modalContent = `
214     <div id="modal" class="modal p-6 border border-slate-200 dark:border-indigo-800 rounded-xl bg-slate-50 dark:bg-slate-800 relative shadow-sm dark:shadow-inner transition-colors duration-300">
215       <div class="flex justify-between items-center mb-4">
216         <div>
217           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Value</label>
218           <input type="text" id="val-modal" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg font-mono text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
219         </div>
220         <div>
221           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Base System</label>
222           <select id="base-modal" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
223           <option value="2">Binary (Base 2)</option>
224           <option value="8">Octal (Base 8)</option>
225           <option value="10" selected>Decimal (Base 10)</option>
226           <option value="16">Hexadecimal (Base 16)</option>
227         </div>
228       </div>
229       <div class="flex justify-between items-center">
230         <div>
231           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Result</label>
232           <div id="result-modal" class="text-slate-900 dark:text-white text-2xl font-mono font-semibold" />
233         </div>
234         <div>
235           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Base</label>
236           <div id="base-modal-label" class="text-slate-900 dark:text-white text-2xl font-mono font-semibold" />
237         </div>
238       </div>
239     </div>`;
240     document.getElementById('modal-content').innerHTML = modalContent;
241     document.getElementById('modal').classList.remove('hidden');
242   }
243
244   function closeModal() {
245     document.getElementById('modal').classList.add('hidden');
246   }
247
248   // Initial Setup for Modal
249   reindexLabels();
250   calculateTotal();
251   updateRemoveButtons();
252   addCase();
253
254   // Close Modal
255   document.getElementById('close-modal').addEventListener('click', closeModal);
256
257   // Show Modal
258   document.getElementById('show-modal').addEventListener('click', showModal);
259
260   // Modal Content
261   function showModal() {
262     const modalContent = `
263     <div id="modal" class="modal p-6 border border-slate-200 dark:border-indigo-800 rounded-xl bg-slate-50 dark:bg-slate-800 relative shadow-sm dark:shadow-inner transition-colors duration-300">
264       <div class="flex justify-between items-center mb-4">
265         <div>
266           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Value</label>
267           <input type="text" id="val-modal" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg font-mono text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
268         </div>
269         <div>
270           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Base System</label>
271           <select id="base-modal" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
272           <option value="2">Binary (Base 2)</option>
273           <option value="8">Octal (Base 8)</option>
274           <option value="10" selected>Decimal (Base 10)</option>
275           <option value="16">Hexadecimal (Base 16)</option>
276         </div>
277       </div>
278       <div class="flex justify-between items-center">
279         <div>
280           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Result</label>
281           <div id="result-modal" class="text-slate-900 dark:text-white text-2xl font-mono font-semibold" />
282         </div>
283         <div>
284           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Base</label>
285           <div id="base-modal-label" class="text-slate-900 dark:text-white text-2xl font-mono font-semibold" />
286         </div>
287       </div>
288     </div>`;
289     document.getElementById('modal-content').innerHTML = modalContent;
290     document.getElementById('modal').classList.remove('hidden');
291   }
292
293   function closeModal() {
294     document.getElementById('modal').classList.add('hidden');
295   }
296
297   // Initial Setup for Modal
298   reindexLabels();
299   calculateTotal();
300   updateRemoveButtons();
301   addCase();
302
303   // Close Modal
304   document.getElementById('close-modal').addEventListener('click', closeModal);
305
306   // Show Modal
307   document.getElementById('show-modal').addEventListener('click', showModal);
308
309   // Modal Content
310   function showModal() {
311     const modalContent = `
312     <div id="modal" class="modal p-6 border border-slate-200 dark:border-indigo-800 rounded-xl bg-slate-50 dark:bg-slate-800 relative shadow-sm dark:shadow-inner transition-colors duration-300">
313       <div class="flex justify-between items-center mb-4">
314         <div>
315           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Value</label>
316           <input type="text" id="val-modal" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg font-mono text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
317         </div>
318         <div>
319           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Base System</label>
320           <select id="base-modal" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
321           <option value="2">Binary (Base 2)</option>
322           <option value="8">Octal (Base 8)</option>
323           <option value="10" selected>Decimal (Base 10)</option>
324           <option value="16">Hexadecimal (Base 16)</option>
325         </div>
326       </div>
327       <div class="flex justify-between items-center">
328         <div>
329           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Result</label>
330           <div id="result-modal" class="text-slate-900 dark:text-white text-2xl font-mono font-semibold" />
331         </div>
332         <div>
333           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Base</label>
334           <div id="base-modal-label" class="text-slate-900 dark:text-white text-2xl font-mono font-semibold" />
335         </div>
336       </div>
337     </div>`;
338     document.getElementById('modal-content').innerHTML = modalContent;
339     document.getElementById('modal').classList.remove('hidden');
340   }
341
342   function closeModal() {
343     document.getElementById('modal').classList.add('hidden');
344   }
345
346   // Initial Setup for Modal
347   reindexLabels();
348   calculateTotal();
349   updateRemoveButtons();
350   addCase();
351
352   // Close Modal
353   document.getElementById('close-modal').addEventListener('click', closeModal);
354
355   // Show Modal
356   document.getElementById('show-modal').addEventListener('click', showModal);
357
358   // Modal Content
359   function showModal() {
360     const modalContent = `
361     <div id="modal" class="modal p-6 border border-slate-200 dark:border-indigo-800 rounded-xl bg-slate-50 dark:bg-slate-800 relative shadow-sm dark:shadow-inner transition-colors duration-300">
362       <div class="flex justify-between items-center mb-4">
363         <div>
364           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Value</label>
365           <input type="text" id="val-modal" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg font-mono text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
366         </div>
367         <div>
368           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Base System</label>
369           <select id="base-modal" class="w-full bg-white dark:bg-slate-900 border border-slate-300 dark:border-indigo-700 p-3 rounded-lg text-lg text-slate-900 dark:text-white focus:outline-none focus:ring-2 focus:ring-indigo-500" />
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376       <div class="flex justify-between items-center">
377         <div>
378           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Result</label>
379           <div id="result-modal" class="text-slate-900 dark:text-white text-2xl font-mono font-semibold" />
380         </div>
381         <div>
382           <label class="block text-sm font-semibold text-slate-600 dark:text-indigo-300 mb-1">Base</label>
383           <div id="base-modal-label" class="text-slate-900 dark:text-white text-2xl font-mono font-semibold" />
384         </div>
385       </div>
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388     document.getElementById('modal').classList.remove('hidden');
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712           &lt
```

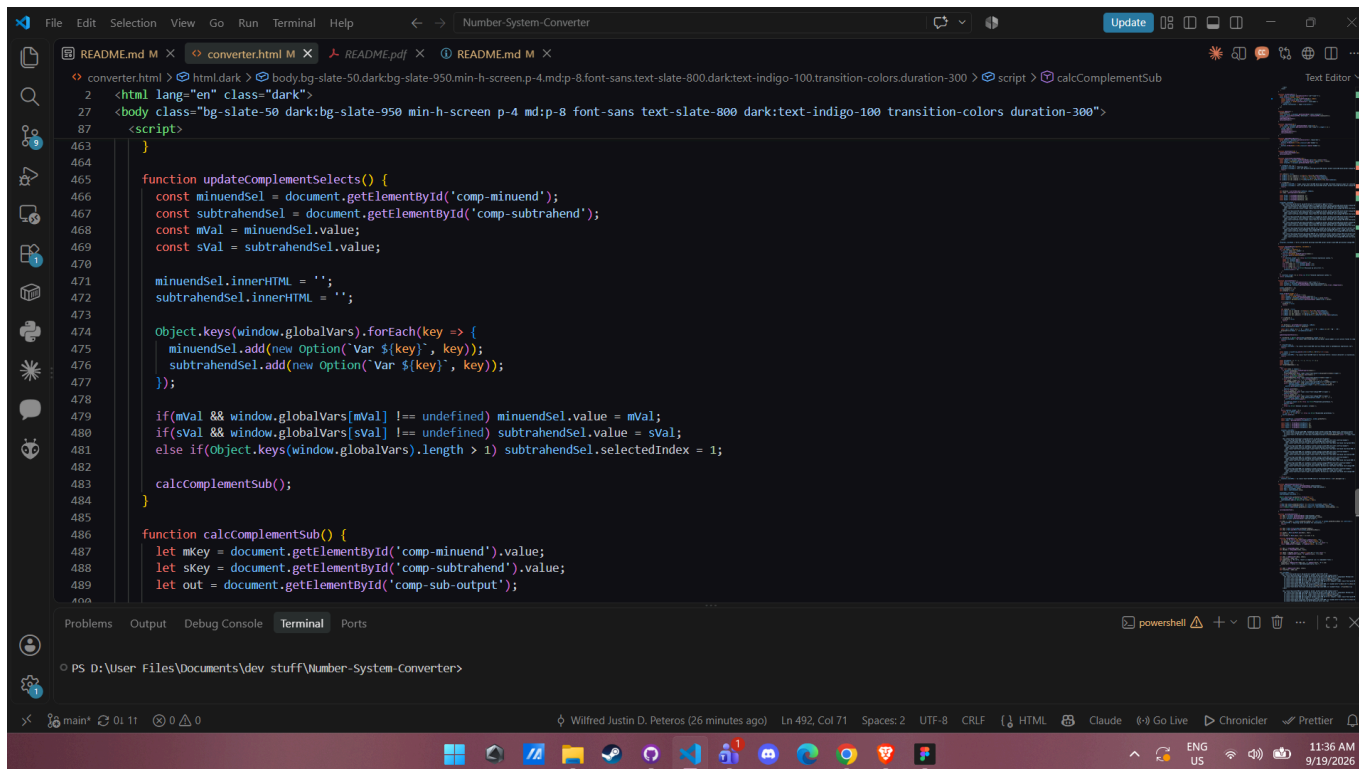
```
File Edit Selection View Go Run Terminal Help
Number-System-Converter > converter.html > html:dark > @script > theme > extend
2 <html lang="en" class="dark">
27 <body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
71 <script>
165 function removeUseless() {
173 }
174
175 function updateRemoveButtons() {
176   const buttons = document.querySelectorAll('.remove-btn');
177   if (buttons.length <= 2) {
178     buttons.forEach(btn => btn.classList.add('hidden'));
179   } else {
180     buttons.forEach(btn => btn.classList.remove('hidden'));
181   }
182 }
183
184 function handleInput(id) {
185   convertIndividualNumber(id);
186   calculateTotal();
187 }
188
189 function convertIndividualNumber(id) {
190   const rawValue = document.getElementById('val-' + id).value.trim();
191   const inBase = parseInt(document.getElementById('inBase-' + id).value);
192   const outputDiv = document.getElementById('out-' + id);
193
194   if (rawValue === "") {
195     outputDiv.innerHTML = "Awaiting input...";
196     outputDiv.className = "mt-4 p-4 bg-white dark:bg-slate-950 border border-slate-200 dark:border-indigo-900 rounded-lg min-h-[80px] flex items-center justify-center text-slate-500 dark:text-indigo-400 text-lg";
197     return;
198   }
199
200   // Regex updated to support optional fractional parts (decimals)
201   let isValid = false;
202   if (inBase === 2) isValid = /^[01]*(\.[01]+)?$/i.test(rawValue);
203   if (inBase === 8) isValid = /^[0-7]*(\.[0-7]+)?$/i.test(rawValue);
204   if (inBase === 10) isValid = /^[0-9]*(\.[0-9]+)?$/i.test(rawValue);
205   if (inBase === 16) isValid = /^[0-9a-fA-F]*(\.[0-9a-fA-F]+)?$/i.test(rawValue);
206
207   if (!isValid) {
208     outputDiv.innerHTML = "<span class='text-red-600 dark:text-red-400 font-bold'>Invalid input for selected base.</span>";
209     outputDiv.className = "mt-4 p-4 bg-white dark:bg-slate-950 border border-slate-200 dark:border-indigo-900 rounded-lg min-h-[80px] flex items-center justify-center transition-colors";
210     return;
211   }
212
213   let decValue = parseToDecimal(rawValue, inBase);
214
215   const binStr = formatBase(decValue, 2);
216   const octStr = formatBase(decValue, 8);
217   const decStr = formatBase(decValue, 10);
218   const hexStr = formatBase(decValue, 16);
219
220   outputDiv.innerHTML = `
221     <div class="grid grid-cols-2 lg:grid-cols-4 gap-3 w-full">
222       <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
223         <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Binary</span>
224         <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${binStr}</span></div>
225       </div>
226       <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
227         <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Octal</span>
228         <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${octStr}</span></div>
229       </div>
230       <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
231         <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Decimal</span>
232         <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${decStr}</span></div>
233       </div>
234       <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
235         <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Hexadecimal</span>
236         <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${hexStr}</span></div>
237       </div>
238     </div>
239   `;
240   outputDiv.className = "mt-4 p-3 bg-white dark:bg-slate-950 border border-slate-200 dark:border-indigo-900 rounded-lg min-h-[80px] transition-colors";
241 }
242
243 function evaluatePostfix(postfix, variables) {
244   const valStack = [];
245   for (let token of postfix) {
246     if (typeof token === 'number') {
247       valStack.push(token);
248     } else if (variables.hasOwnProperty(token)) {
249       valStack.push(variables[token]);
250     } else {
251       if (valStack.length < 2) throw new Error("Invalid expression syntax.");
252       const b = valStack.pop();
253       const a = valStack.pop();
254       if (token === '+') valStack.push(a + b);
255       else if (token === '-') valStack.push(a - b);
256       else if (token === '*') valStack.push(a * b);
257       else if (token === '/') {
258         if (b === 0) throw new Error("Division by zero error.");
259       }
260     }
261   }
262   return valStack[0];
263 }
```

```
File Edit Selection View Go Run Terminal Help
Number-System-Converter > converter.html > html:dark > @script > theme > extend
2 <html lang="en" class="dark">
27 <body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
71 <script>
189 function convertIndividualNumber(id) {
216   const octStr = formatBase(decValue, 8);
217   const decStr = formatBase(decValue, 10);
218   const hexStr = formatBase(decValue, 16);
219
220   outputDiv.innerHTML = `
221     <div class="grid grid-cols-2 lg:grid-cols-4 gap-3 w-full">
222       <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
223         <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Binary</span>
224         <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${binStr}</span></div>
225       </div>
226       <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
227         <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Octal</span>
228         <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${octStr}</span></div>
229       </div>
230       <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
231         <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Decimal</span>
232         <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${decStr}</span></div>
233       </div>
234       <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
235         <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Hexadecimal</span>
236         <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${hexStr}</span></div>
237       </div>
238     </div>
239   `;
240   outputDiv.className = "mt-4 p-3 bg-white dark:bg-slate-950 border border-slate-200 dark:border-indigo-900 rounded-lg min-h-[80px] transition-colors";
241 }
242
243 function evaluatePostfix(postfix, variables) {
244   const valStack = [];
245   for (let token of postfix) {
246     if (typeof token === 'number') {
247       valStack.push(token);
248     } else if (variables.hasOwnProperty(token)) {
249       valStack.push(variables[token]);
250     } else {
251       if (valStack.length < 2) throw new Error("Invalid expression syntax.");
252       const b = valStack.pop();
253       const a = valStack.pop();
254       if (token === '+') valStack.push(a + b);
255       else if (token === '-') valStack.push(a - b);
256       else if (token === '*') valStack.push(a * b);
257       else if (token === '/') {
258         if (b === 0) throw new Error("Division by zero error.");
259       }
260     }
261   }
262   return valStack[0];
263 }
```

```
Number-System-Converter > converter.html > html:dark > head > script > theme > extend
2 <html lang="en" class="dark">
27 <body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
71 <script>
243 function evaluatePostfix(postfix, variables) {
260 }
261 }
262 }
263 if (valStack.length !== 1) throw new Error("Invalid expression syntax.");
264 return valStack[0];
265 }
266
267 function calculateTotal() {
268 const cases = document.querySelectorAll('[id~="case-"]');
269 const totalDiv = document.getElementById('final-total-container');
270 const exprString = document.getElementById('math-expression').value.trim().toLowerCase();
271
272 let variables = {};
273 let displayStrs = {};
274 let allValid = true;
275
276 cases.forEach(caseEl => {
277 const id = caseEl.id.split('-')[1];
278 const varName = caseEl.getAttribute('data-var');
279 const rawValue = document.getElementById(`val-${id}`).value.trim();
280 const inBase = parseInt(document.getElementById(`inBase-${id}`).value);
281
282 if (!rawValue) {
283 allValid = false;
284 return;
285 }
286
287 let isValid = false;
288 if (inBase === 2) isValid = /^[01]+$/i.test(rawValue);
289 if (inBase === 8) isValid = /^[0-7]+$/i.test(rawValue);
290 if (inBase === 10) isValid = /^[0-9]+$/i.test(rawValue);
291 if (inBase === 16) isValid = /^[0-9a-fA-F]+$/i.test(rawValue);
292
293 if (!isValid) {
294 allValid = false;
295 return;
296 }
297
298 let decValue = parseDecimal(rawValue, inBase);
299 variables[varName] = decValue;
300
301 const sub = inBase === 2 ? 'b' : inBase === 8 ? 'o' : inBase === 10 ? 'd' : 'x';
302 displayStrs[varName] = `(${rawValue})${sub}`;
303 });
304
305 if (!allValid || Object.keys(variables).length === 0) {
306 totalDiv.innerHTML = "<p class='text-slate-400 text-lg'>Enter valid numbers in all active fields to compute.</p>";
307 return;
308 }
309
310 if (!exprString) {
311 totalDiv.innerHTML = "<p class='text-slate-400 text-lg'>Please enter a mathematical expression.</p>";
312 return;
313 }
314
315 // Regex updated to parse direct decimal numbers typed in the expression field
316 const tokens = exprString.match(/([A-Z])|([0-9]*\.\?[0-9]*)|([+\-*/()])|([0-9])/g);
317 if (!tokens) {
318 totalDiv.innerHTML = "<p class='text-red-400 text-xl font-bold'>Error: Invalid characters in expression.</p>";
319 return;
320 }
321
322 const precedence = { '+': 1, '-': 1, '*': 2, '/': 2 };
323 const postfix = [];
324 const opStack = [];
325 let formattedEqTokens = [];
326
327 try {
328 for (let token of tokens) {
329 if (variables.hasOwnProperty(token)) {
330 postfix.push(token);
331 formattedEqTokens.push("<span class='text-white'>${displayStrs[token]}</span>");
332 } else if (/^[0-9]*\.\?[0-9]*$/i.test(token)) {
333 postfix.push(parseFloat(token));
334 formattedEqTokens.push("<span class='text-white'>${token}</span>");
335 } else if (!['+', '-', '*', '/'].includes(token)) {
336 let symbol = token === '*' ? '×' : token === '/' ? '÷' : token;
337 formattedEqTokens.push("<span class='text-pink-400 mx-1'>${symbol}</span>");
338 while (opStack.length > 0 && opStack[opStack.length - 1] !== '(' &&
339 precedence[opStack[opStack.length - 1]] >= precedence[token]) {
340 postfix.push(opStack.pop());
341 }
342 opStack.push(token);
343 } else if (token === '(') {
344 formattedEqTokens.push("<span class='text-indigo-400'>(</span>");
345 opStack.push(token);
346 }
347 }
348 while (opStack.length > 0) {
349 postfix.push(opStack.pop());
350 }
351 formattedEqTokens.push("<span class='text-white'>= </span>");
352 } catch {
353 totalDiv.innerHTML = "<p class='text-red-400 text-xl font-bold'>Error: Invalid expression.</p>";
354 return;
355 }
356
357 totalDiv.innerHTML = formattedEqTokens.join('');
358 }
```

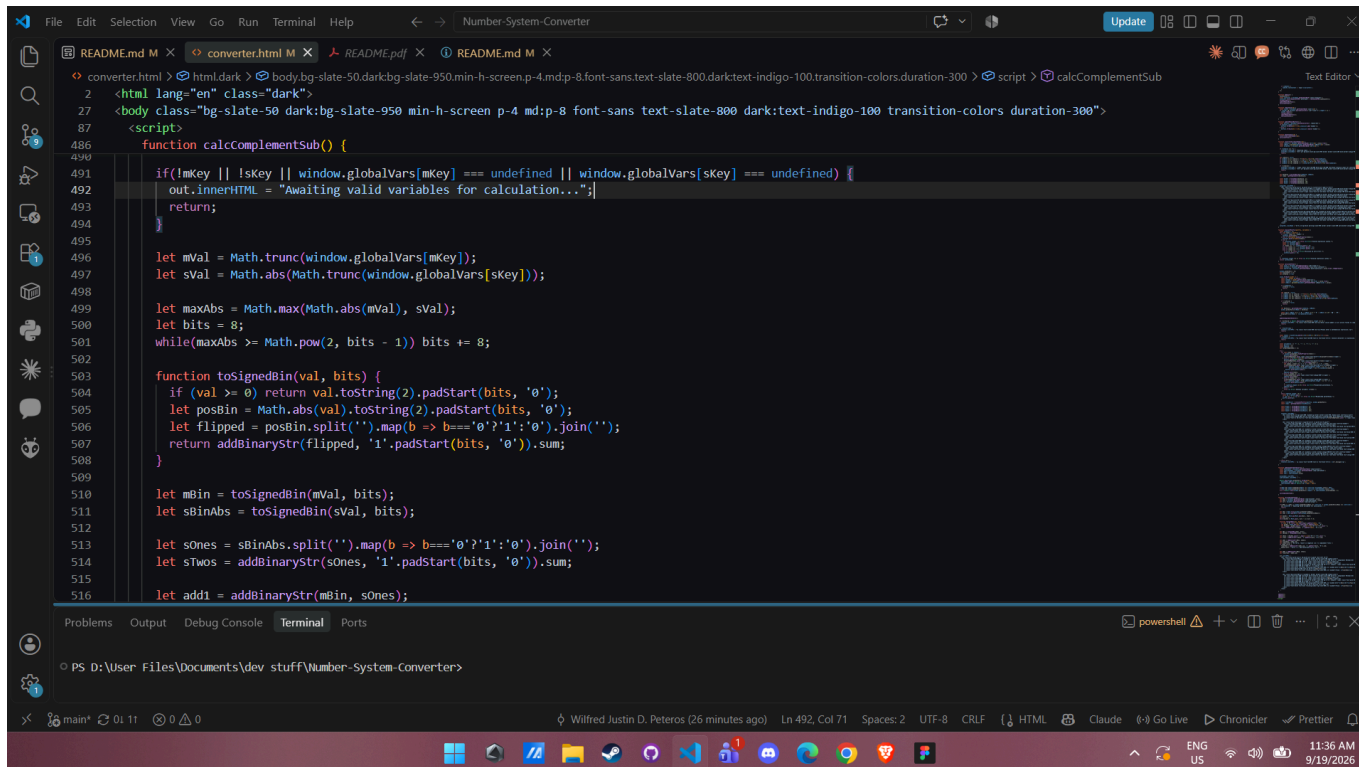
```
Number-System-Converter > converter.html > html:dark > head > script > theme > extend
2 <html lang="en" class="dark">
27 <body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
71 <script>
267 function calculateTotal() {
342   opStack.push(token);
343 } else if (token === '(') {
344   formattedTokens.push('<span class="text-indigo-400"></span>');
345   opStack.push(token);
346 } else if (token === ')') {
347   formattedTokens.push('<span class="text-indigo-400"></span>');
348   while (opStack.length > 0 && opStack[opStack.length - 1] !== '(') {
349     postfix.push(opStack.pop());
350   }
351   if (opStack.length === 0) throw new Error("Mismatched parentheses.");
352   opStack.pop();
353 } else {
354   throw new Error("Unknown variable: ${token}");
355 }
356 }
357 while (opStack.length > 0) {
358   const op = opStack.pop();
359   if (op === '(' || op === ')') throw new Error("Mismatched parentheses.");
360   postfix.push(op);
361 }
362
363 const finalResult = evaluatePostfix(postfix, variables);
364
365 const binStr = formatBase(finalResult, 2);
366 const octStr = formatBase(finalResult, 8);
367 const decStr = formatBase(finalResult, 10);
368 const hexStr = formatBase(finalResult, 16);
369
370 totalDiv.innerHTML = `
371 <div class="mb-6 p-4 bg-slate-800 rounded-lg border border-slate-700 shadow-inner overflow-x-auto">
372   <p class="text-indigo-300 text-sm font-bold uppercase tracking-wider mb-2">Equation Breakdown</p>
373   <p class="text-xl md:text-2xl font-mono whitespace-normal">${formattedEqTokens.join('')} <span class="text-pink-400 mx-2">=</span></p>
374 </div>
375
376 <div class="grid grid-cols-1 md:grid-cols-2 gap-4">
377   <div class="bg-slate-800 p-4 rounded-xl border border-slate-700 text-left overflow-hidden">
378     <span class="block text-sm font-bold text-slate-400 mb-1">Binary</span>
379     <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-mono font-bold text-green-400 break-all">${binStr}</span></div>
380   </div>
381   <div class="bg-slate-800 p-4 rounded-xl border border-slate-700 text-left overflow-hidden">
382     <span class="block text-sm font-bold text-slate-400 mb-1">Octal</span>
383     <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-mono font-bold text-blue-400 break-all">${octStr}</span></div>
384   </div>
385   <div class="bg-slate-800 p-4 rounded-xl border border-slate-700 text-left overflow-hidden">
386     <span class="block text-sm font-bold text-slate-400 mb-1">Decimal</span>
387     <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-mono font-bold text-yellow-400 break-all">${decStr}</span></div>
388   </div>
389   <div class="bg-slate-800 p-4 rounded-xl border border-slate-700 text-left overflow-hidden">
390     <span class="block text-sm font-bold text-slate-400 mb-1">Hexadecimal</span>
391     <div class="overflow-x-auto"><span class="text-xl md:text-2xl font-mono font-bold text-pink-400 break-all">${hexStr}</span></div>
392   </div>
393 </div>
394
395 }
396 } catch (err) {
397   totalDiv.innerHTML = `<p class="text-red-400 text-xl font-bold">Error: ${err.message}</p>`;
398 }
399
400 addCase();
401 addCase();
402 addCase();
403 </script>
404 </body>
405 </html>
```

```
File Edit Selection View Go Run Terminal Help Number-System-Converter Update
converter.html X converter.html X README.pdf X README.mmd X ss-11.png X
converter.html > <html lang="en" class="dark">
2 <body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
27 <script>
87
122
123
124 function getComplements(num) {
125   if (isNaN(num)) return { ones: "NaN", twos: "NaN" };
126   let intVal = Math.trunc(Math.abs(num));
127   let binStr = intVal.toString(2);
128
129   let bits = 8;
130   while (binStr.length > bits - 1) bits += 8;
131   binStr = binStr.padStart(bits, '0');
132
133   let onesComp = binStr.split('').map(b => b === '0' ? '1' : '0').join('');
134
135   let twosComp = '';
136   let carry = 1;
137   for (let i = bits - 1; i >= 0; i--) {
138     let sum = parseInt(onesComp[i]) + carry;
139     twosComp = (sum % 2) + twosComp;
140     carry = Math.floor(sum / 2);
141   }
142
143   return { ones: onesComp, twos: twosComp, bits: bits };
144 }
145
146 function addBinaryStr(a, b) {
147   let res = '', carry = 0;
148   for (let i = a.length - 1; i >= 0; i--) {
149
150   }
151 }
152
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```



This screenshot shows the VS Code editor with the file `converter.html` open. The editor displays the `updateComplementSelects` function, which updates the innerHTML of two select elements based on the values of `mVal` and `sVal`. The function also updates the `selectedIndex` of the `subtrahendSel` element. The `calcComplementSub` function is also visible, which calculates the complement of the input values.

```
2 <html lang="en" class="dark">
27 <body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
87 <script>
463
464
465 function updateComplementSelects() {
466   const minuendSel = document.getElementById('comp-minuend');
467   const subtrahendSel = document.getElementById('comp-subtrahend');
468   const mVal = minuendSel.value;
469   const sVal = subtrahendSel.value;
470
471   minuendSel.innerHTML = '';
472   subtrahendSel.innerHTML = '';
473
474   Object.keys(window.globalVars).forEach(key => {
475     minuendSel.add(new Option('Var ${key}', key));
476     subtrahendSel.add(new Option('Var ${key}', key));
477   });
478
479   if(mVal && window.globalVars[mVal] !== undefined) minuendSel.value = mVal;
480   if(sVal && window.globalVars[sVal] !== undefined) subtrahendSel.value = sVal;
481   else if(Object.keys(window.globalVars).length > 1) subtrahendSel.selectedIndex = 1;
482
483   calcComplementSub();
484 }
485
486 function calcComplementSub() {
487   let mKey = document.getElementById('comp-minuend').value;
488   let sKey = document.getElementById('comp-subtrahend').value;
489   let out = document.getElementById('comp-sub-output');
```



This screenshot shows the VS Code editor with the file `converter.html` open. The editor displays the `calcComplementSub` function, which calculates the complement of the input values. The function uses `Math.trunc` to truncate the input values and `Math.abs` to get the absolute value. It then uses `Math.max` to get the maximum of the two values and `Math.pow` to calculate the number of bits. The function also uses `toSignedBin` to convert the input values to signed binary and `addBinaryStr` to add the two binary strings.

```
486 function calcComplementSub() {
490   if(!mKey || !sKey || window.globalVars[mKey] === undefined || window.globalVars[sKey] === undefined) {
491     out.innerHTML = "Awaiting valid variables for calculation...";
492     return;
493   }
494
495   let mVal = Math.trunc(window.globalVars[mKey]);
496   let sVal = Math.abs(Math.trunc(window.globalVars[sKey]));
497
498   let maxAbs = Math.max(Math.abs(mVal), sVal);
499   let bits = 8;
500   while(maxAbs >= Math.pow(2, bits - 1)) bits += 8;
501
502   function toSignedBin(val, bits) {
503     if (val >= 0) return val.toString(2).padStart(bits, '0');
504     let posBin = Math.abs(val).toString(2).padStart(bits, '0');
505     let flipped = posBin.split('').map(b => b==='0'? '1': '0').join('');
506     return addBinaryStr(flipped, '1'.padStart(bits, '0')).sum;
507   }
508
509   let mBin = toSignedBin(mVal, bits);
510   let sBinAbs = toSignedBin(sVal, bits);
511
512   let sOnes = sBinAbs.split('').map(b => b==='0'? '1': '0').join('');
513   let sTwos = addBinaryStr(sOnes, '1'.padStart(bits, '0')).sum;
514
515   let add1 = addBinaryStr(mBin, sOnes);
516 }
```

```
File Edit Selection View Go Run Terminal Help Number-System-Converter Update
converter.html X html.dar X body.bg-slate-50.dark.bg-slate-950.min-h-screen.p-4.md.p-8.font-sans.text-slate-800.dark:text-indigo-100.transition-colors.duration-300 X script X calcComplementSub
2 <html lang="en" class="dark">
27 <body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
87 <script>
486 function calcComplementSub() {
514 let sTwos = addBinaryStr(sOnes, '1'.padStart(bits, '0')).sum;
515
516 let add1 = addBinaryStr(mBin, sOnes);
517 let finalOnes = add1.sum;
518 let endCarryTxt = "No carry, result is negative (in 1's complement form).";
519 if (add1.carry) {
520 finalOnes = addBinaryStr(add1.sum, '1'.padStart(bits, '0')).sum;
521 endCarryTxt = "Carry = 1. Add End-Around Carry (+1).";
522 }
523
524 let add2 = addBinaryStr(mBin, sTwos);
525 let finalTwos = add2.sum;
526
527 out.innerHTML = `
528 <div class="grid grid-cols-1 md:grid-cols-2 gap-8 text-left mt-4">
529 <div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner">
530 <h3 class="text-teal-400 font-bold mb-4 border-b border-slate-700 pb-2">1's Complement Method</h3>
531 <p class="text-slate-400 mb-1">M: <span class="text-white">${mBin}</span></p>
532 <p class="text-slate-400 mb-1">S: <span class="text-white">${sBinAbs}</span></p>
533 <p class="text-slate-400 mb-3 border-b border-slate-700 pb-2">1's Comp(S): <span class="text-pink-400">${sOnes}</span></p>
534 <p class="text-slate-400 mb-1">Add M + 1's Comp(S):</p>
535 <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded mb-2">${mBin}<br>+${sOnes}<br>-----<br>${add1.carry ? '1' : '0'}${add1.sum}<br>
536 <p class="text-emerald-400 text-sm mb-2">${endCarryTxt}</p>
537 <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded">Final: ${finalOnes}</p>
538 </div>
539 <div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner">
540 </div>
541 </div>
542 `;
543 }
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545 addCase();
546 addCase();
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```
File Edit Selection View Go Run Terminal Help Number-System-Converter Update
README.md M X converter.html M X README.pdf X README.md M X
ion-colors.duration-300 > div.mt-8.p-8.bg-slate-800.dark:bg-slate-950.border-2.border-emerald-500.dark:border-emerald-600.rounded-2xl.shadow-xl.relative.overflow-hidden > div#comp-sub-output.text-slate-300.text-center.font-mono Text Editor
2 m1 lang="en" class="dark">
27 dy class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
29 div class="max-w-6xl mx-auto">
41 <div class="bg-white dark:bg-slate-900 p-6 md:p-8 rounded-2xl shadow-xl border border-slate-200 dark:border-indigo-900 transition-colors duration-300">
60 <div class="mt-12 p-8 bg-slate-900 dark:bg-black border-2 border-indigo-500 dark:border-purple-600 rounded-2xl shadow-2xl relative overflow-hidden">
65 </div>
66 </div>
67
68 <div class="mt-8 p-8 bg-slate-800 dark:bg-slate-950 border-2 border-emerald-500 dark:border-emerald-600 rounded-2xl shadow-xl relative overflow-hidden">
69 <div class="absolute top-0 left-0 w-full h-1 bg-gradient-to-r from-emerald-400 to-teal-500"></div>
70 <h2 class="text-2xl font-bold text-white mb-2 text-center">Complement Subtraction Method</h2>
71 <p class="text-sm text-center text-emerald-400 mb-6">Select two variables to demonstrate step-by-step subtraction (Minuend - Subtrahend) using complements. Uses abs
72
73 <div class="flex justify-center items-center gap-4 mb-6">
74 <select id="comp-minuend" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-emerald-500 font-bold text-lg" onchange="calcComplementSub()"></select>
75 <span class="text-white text-3xl font-bold">-</span>
76 <select id="comp-subtrahend" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-emerald-500 font-bold text-lg" onchange="calcComplementSub()"></select>
77 </div>
78
79 <div id="comp-sub-output" class="text-slate-300 text-center font-mono">
82 | Awaiting variable selection...
83 </div>
84 </div>
85
86 </div>
87 /div>
88
89 script>
Problems Output Debug Console Terminal Ports powershell
PS D:\User Files\Documents\dev stuff\Number-System-Converter>
main* 01.11 0 0 0 Wilfred Justin D. Peters (28 minutes ago) Ln 83, Col 15 Spaces: 2 UTF-8 CRLF HTML Claude Go Live Chronider Prettier 11:37 AM 9/19/2026
```

```
File Edit Selection View Go Run Terminal Help Number-System-Converter Update
converter.html M X converter.html M X README.pdf X README.md M X
converter.html > html.dark > body.bg-slate-50.dark:bg-slate-950.min-h-screen.p-4.md:p-8.font-sans.text-slate-800.dark:text-indigo-100.transition-colors.duration-300 > script > evaluatePostfix
2 <html lang="en" class="dark">
27 <body class="bg-slate-50 dark:bg-slate-950 min-h-screen p-4 md:p-8 font-sans text-slate-800 dark:text-indigo-100 transition-colors duration-300">
89 <script>
238 function convertIndividualNumber(id) {
239 <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
280 <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Decimal</span>
281 <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${decStr}</span></div>
282 </div>
283 <div class="bg-slate-50 dark:bg-slate-900 p-3 rounded-lg border border-slate-200 dark:border-indigo-800 min-w-0 flex flex-col">
284 <span class="text-xs font-semibold text-slate-500 dark:text-indigo-400 mb-1">Hexadecimal</span>
285 <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-700 dark:text-purple-300 break-all">${hexStr}</span></div>
286 </div>
287 <div class="bg-indigo-50 dark:bg-indigo-950/40 p-3 rounded-lg border border-indigo-200 dark:border-indigo-800 min-w-0 flex flex-col">
288 <span class="text-xs font-semibold text-indigo-600 dark:text-indigo-400 mb-1">1's Comp (Int)</span>
289 <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-800 dark:text-purple-200 break-all">${comps.ones}</span></div>
290 </div>
291 <div class="bg-indigo-50 dark:bg-indigo-950/40 p-3 rounded-lg border border-indigo-200 dark:border-indigo-800 min-w-0 flex flex-col">
292 <span class="text-xs font-semibold text-indigo-600 dark:text-indigo-400 mb-1">2's Comp (Int)</span>
293 <div class="overflow-x-auto"><span class="text-sm font-mono font-bold text-indigo-800 dark:text-purple-200 break-all">${comps.twos}</span></div>
294 </div>
295 </div>
296
297 outputDiv.className = "mt-4 p-3 bg-white dark:bg-slate-950 border border-slate-200 dark:border-indigo-900 rounded-lg min-h-[80px] transition-colors";
298 }
299
300 function evaluatePostfix(postfix, variables) {
301 const valStack = [];
302 for (let token of postfix) {
303 if (typeof token === 'number') {
304 valStack.push(token);
305 } else if (variables.hasOwnProperty(token)) {
...
Problems Output Debug Console Terminal Ports powershell
PS D:\User Files\Documents\dev stuff\Number-System-Converter>
main* 01.11 0 0 0 LAB64_USER ON WS41 (1 week ago) Ln 302, Col 35 Spaces: 2 UTF-8 CRLF HTML Claude Go Live Chronider Prettier 11:38 AM 9/19/2026
```

```
File Edit Selection View Go Run Terminal Help
converter.html X README.md X
1 # Activity No. 1: Number System Converter and Arithmetic Calculator
183 ## Program Implementation
204 <div id="comp-sub-output" class="text-slate-300 text-center font-mono">
205   Awaiting variable selection...
206 </div>
207
208 <div class="mt-8 p-8 bg-slate-800 dark:bg-slate-950 border-2 border-blue-500 dark:border-blue-600 rounded-2xl shadow-xl relative overflow-hidden">
209   <div class="absolute top-0 left-0 w-full h-1 bg-gradient-to-r from-blue-400 to-cyan-500"></div>
210   <h2 class="text-2xl font-bold text-white mb-2 text-center">BCD Addition & Subtraction</h2>
211   <p class="text-sm text-center text-blue-400 mb-6">Select two variables and an operation. Subtraction displays both 9's and 10's complement methods.</p>
212
213   <div class="flex justify-center items-center gap-4 mb-6">
214     <select id="bcd-var1" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-blue-500 font-bold text-lg"
215       onchange="calcBCD()"></select>
216     <select id="bcd-op" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-blue-500 font-bold text-lg"
217       onchange="calcBCD()">
218       <option value="+">+</option>
219       <option value="-">-</option>
220     </select>
221     <select id="bcd-var2" class="bg-slate-700 text-white p-3 rounded-lg border border-slate-600 focus:ring-2 focus:ring-blue-500 font-bold text-lg"
222       onchange="calcBCD()"></select>
223   </div>
224
225   <div id="bcd-output" class="text-slate-300 text-center font-mono">
226     Awaiting variable selection...
227   </div>
228 </div>
229 </div>
230 <script>
```

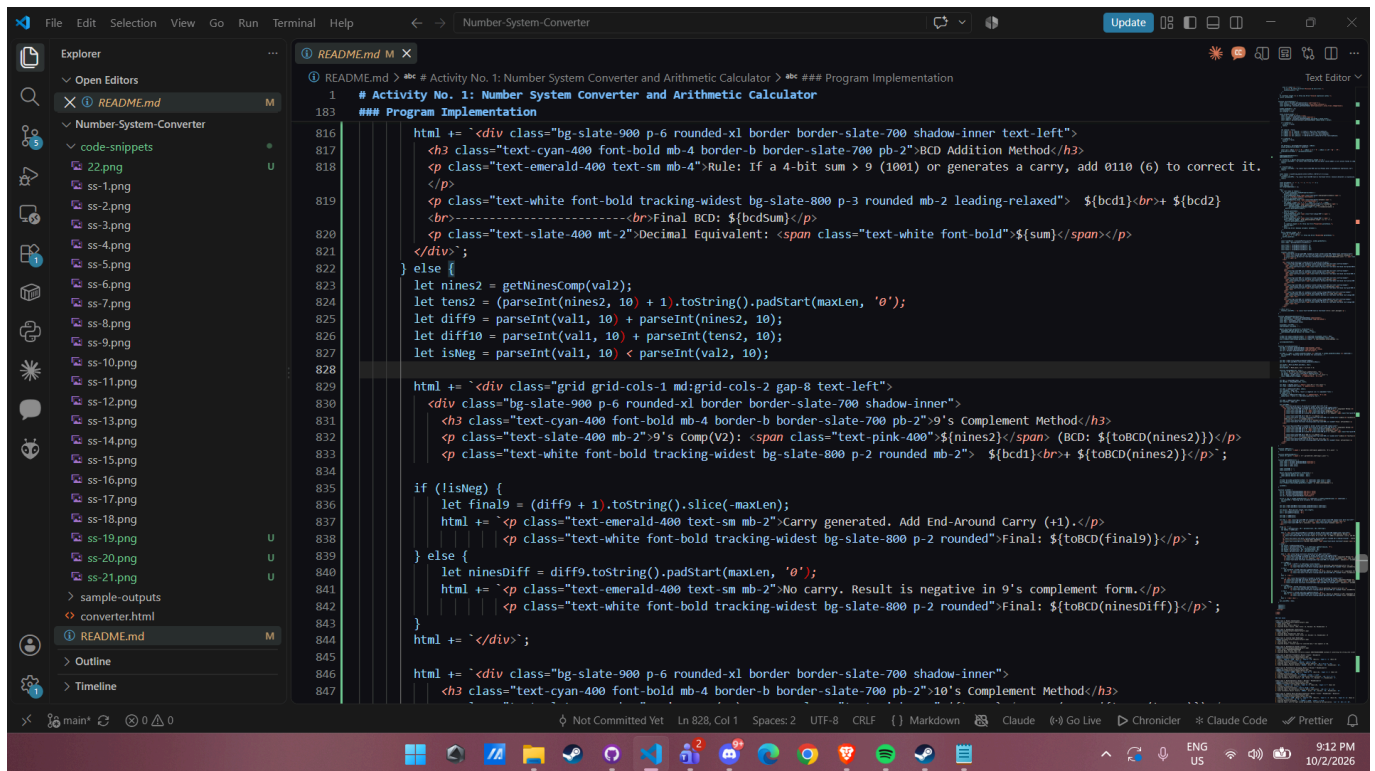
```
File Edit Selection View Go Run Terminal Help
converter.html X README.md X
1 # Activity No. 1: Number System Converter and Arithmetic Calculator
183 ## Program Implementation
757 function toBCD(str) {
758   return str.split('').map(d => parseInt(d).toString(2).padStart(4, '0')).join(' ');
759 }
760
761 function getNinesComp(str) {
762   return str.split('').map(d => (9 - parseInt(d)).toString()).join('');
763 }
764
765 function updateBCDSelects() {
766   const v1Sel = document.getElementById('bcd-var1');
767   const v2Sel = document.getElementById('bcd-var2');
768   const v1Val = v1Sel.value;
769   const v2Val = v2Sel.value;
770
771   v1Sel.innerHTML = '';
772   v2Sel.innerHTML = '';
773
774   Object.keys(window.globalVars).forEach(key => {
775     v1Sel.add(new Option(`Var ${key}`, key));
776     v2Sel.add(new Option(`Var ${key}`, key));
777   });
778
779   if (v1Val && window.globalVars[v1Val] !== undefined) v1Sel.value = v1Val;
780   if (v2Val && window.globalVars[v2Val] !== undefined) v2Sel.value = v2Val;
781   else if (Object.keys(window.globalVars).length > 1) v2Sel.selectedIndex = 1;
782
783   calcBCD();
784 }
785
786 function calcBCD() {
```

```
File Edit Selection View Go Run Terminal Help Number-System-Converter Update
README.md x
# Activity No. 1: Number System Converter and Arithmetic Calculator
### Program Implementation

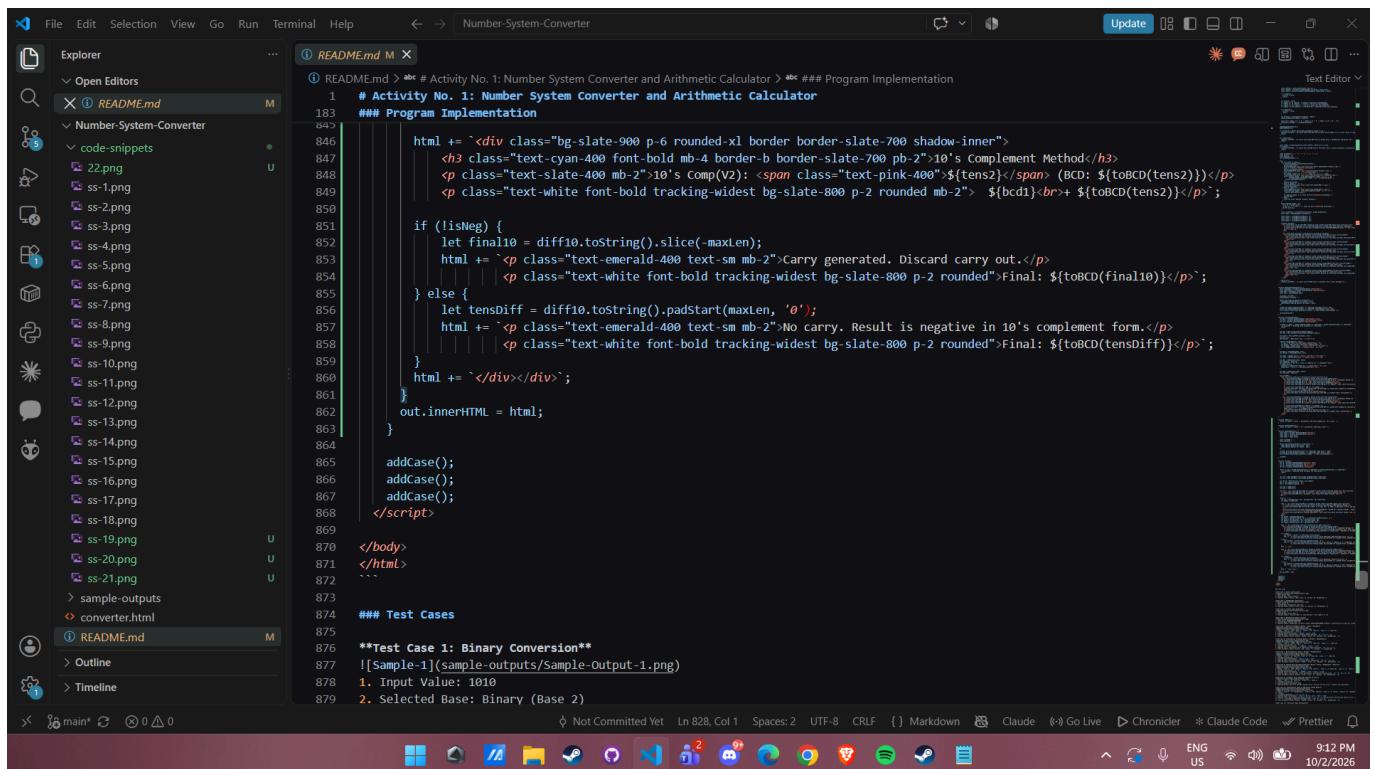
778
779 if (v1Val && window.globalVars[v1Val] !== undefined) v1Sel.value = v1Val;
780 if (v2Val && window.globalVars[v2Val] !== undefined) v2Sel.value = v2Val;
781 else if (Object.keys(window.globalVars).length > 1) v2Sel.selectedIndex = 1;
782
783 calcBCD();
784
785
786 function calcBCD() {
787   let k1 = document.getElementById('bcd-var1').value;
788   let k2 = document.getElementById('bcd-var2').value;
789   let op = document.getElementById('bcd-op').value;
790   let out = document.getElementById('bcd-output');
791
792   if (!k1 || !k2 || window.globalVars[k1] === undefined || window.globalVars[k2] === undefined) {
793     out.innerHTML = "Awaiting valid variables for calculation...";
794     return;
795   }
796
797   let val1 = Math.abs(Math.trunc(window.globalVars[k1])).toString();
798   let val2 = Math.abs(Math.trunc(window.globalVars[k2])).toString();
799
800   let maxLen = Math.max(val1.length, val2.length);
801   val1 = val1.padStart(maxLen, '0');
802   val2 = val2.padStart(maxLen, '0');
803
804   let bcd1 = toBCD(val1);
805   let bcd2 = toBCD(val2);
806
807   let html = `<div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner mb-4 text-left">
808     <p class="text-slate-400 mb-1">Var 1 (${val1}): <span class="text-white">${bcd1}</span></p>
809     <p class="text-slate-400">Var 2 (${val2}): <span class="text-white">${bcd2}</span></p>
810   </div>`;
811
```

```
File Edit Selection View Go Run Terminal Help Number-System-Converter Update
README.md x
# Activity No. 1: Number System Converter and Arithmetic Calculator
### Program Implementation

804 let bcd1 = toBCD(val1);
805 let bcd2 = toBCD(val2);
806
807 let html = `<div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner mb-4 text-left">
808   <p class="text-slate-400 mb-1">Var 1 (${val1}): <span class="text-white">${bcd1}</span></p>
809   <p class="text-slate-400">Var 2 (${val2}): <span class="text-white">${bcd2}</span></p>
810 </div>`;
811
812 if (op === '+') {
813   let sum = (parseInt(val1, 10) + parseInt(val2, 10)).toString();
814   let bcdSum = toBCD(sum);
815
816   html += `<div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner text-left">
817     <h3 class="text-cyan-400 font-bold mb-4 border-b border-slate-700 pb-2">BCD Addition Method</h3>
818     <p class="text-emerald-400 text-sm mb-4">Rule: If a 4-bit sum > 9 (1001) or generates a carry, add 0110 (6) to correct it.</p>
819     <p class="text-white font-bold tracking-widest bg-slate-800 p-3 rounded mb-2 leading-relaxed"> ${bcd1}<br>+ ${bcd2}<br>-----<br>Final
820     BCD: ${bcdSum}</p>
821     <p class="text-slate-400 mt-2">Decimal Equivalent: <span class="text-white font-bold">${sum}</span></p>
822   </div>`;
823 } else {
824   let nines2 = getNinesComp(val2);
825   let tens2 = (parseInt(nines2, 10) + 1).toString().padStart(maxLen, '0');
826   let diff9 = parseInt(val1, 10) + parseInt(nines2, 10);
827   let diff10 = parseInt(val1, 10) + parseInt(tens2, 10);
828   let isNeg = parseInt(val1, 10) < parseInt(val2, 10);
829
830   html += `<div class="grid grid-cols-1 md:grid-cols-2 gap-8 text-left">
831     <div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner">
832       <h3 class="text-cyan-400 font-bold mb-4 border-b border-slate-700 pb-2">9's Complement Method</h3>
833       <p class="text-slate-400 mb-2">9's Comp(V2): <span class="text-pink-400">${nines2}</span> (BCD: ${toBCD(nines2)})</p>
834       <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded mb-2"> ${bcd1}<br>+ ${toBCD(nines2)}</p>
835     </div>
836     <div>
837       <h3 class="text-cyan-400 font-bold mb-4 border-b border-slate-700 pb-2">10's Complement Method</h3>
838       <p class="text-slate-400 mb-2">10's Comp(V2): <span class="text-pink-400">${tens2}</span> (BCD: ${toBCD(tens2)})</p>
839       <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded mb-2"> ${bcd1}<br>+ ${toBCD(tens2)}</p>
840     </div>
841   </div>`;
842 }
843
844 if (isNeg) {
845   let final9 = (diff9 + 1).toString().slice(-maxLen);
846   html += `<div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner mb-4 text-left">
847     <p class="text-slate-400 mb-1">Final Result: <span class="text-white">${final9}</span></p>
848   </div>`;
849 }
850
```



```
1 # Activity No. 1: Number System Converter and Arithmetic Calculator
183 ### Program Implementation
816
817 html += `<div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner text-left">
818   <h3 class="text-cyan-400 font-bold mb-4 border-b border-slate-700 pb-2">BCD Addition Method</h3>
819   <p class="text-emerald-400 text-sm mb-4">Rule: If a 4-bit sum > 9 (1001) or generates a carry, add 0110 (6) to correct it.
820   </p>
821   <p class="text-white font-bold tracking-widest bg-slate-800 p-3 rounded mb-2 leading-relaxed"> ${bcd1}<br>+ ${bcd2}
822   <br>-----<br>Final BCD: ${bcdSum}</p>
823   <p class="text-slate-400 mt-2">Decimal Equivalent: <span class="text-white font-bold">${sum}</span></p>
824   </div>;
825 } else {
826   let nines2 = getNinesComp(val2);
827   let tens2 = (parseInt(nines2, 10) + 1).toString().padStart(maxLen, '0');
828   let diff9 = parseInt(val1, 10) + parseInt(nines2, 10);
829   let diff10 = parseInt(val1, 10) + parseInt(tens2, 10);
830   let isNeg = parseInt(val1, 10) < parseInt(val2, 10);
831
832   html += `<div class="grid grid-cols-1 md:grid-cols-2 gap-8 text-left">
833     <div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner">
834       <h3 class="text-cyan-400 font-bold mb-4 border-b border-slate-700 pb-2">9's Complement Method</h3>
835       <p class="text-slate-400 mb-2">9's Comp(V2): <span class="text-pink-400">${nines2}</span> (BCD: ${toBCD(nines2)})</p>
836       <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded mb-2"> ${bcd1}<br>+ ${toBCD(nines2)}</p>
837     </div>
838     <div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner">
839       <h3 class="text-cyan-400 font-bold mb-4 border-b border-slate-700 pb-2">10's Complement Method</h3>
840       <p class="text-emerald-400 text-sm mb-2">Carry generated. Add End-Around Carry (+1)</p>
841       <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded">Final: ${toBCD(final9)}</p>
842     </div>
843   </div>
844   if (!isNeg) {
845     let final9 = (diff9 + 1).toString().slice(-maxLen);
846     html += `<p class="text-emerald-400 text-sm mb-2">Carry generated. Add End-Around Carry (+1)</p>
847     <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded">Final: ${toBCD(final9)}</p>`;
848   } else {
849     let ninesDiff = diff9.toString().padStart(maxLen, '0');
850     html += `<p class="text-emerald-400 text-sm mb-2">No carry. Result is negative in 9's complement form.</p>
851     <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded">Final: ${toBCD(ninesDiff)}</p>`;
852   }
853   html += `</div>`;
854
855   html += `<div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner">
856     <h3 class="text-cyan-400 font-bold mb-4 border-b border-slate-700 pb-2">10's Complement Method</h3>
```



```
846   <div class="bg-slate-900 p-6 rounded-xl border border-slate-700 shadow-inner">
847     <h3 class="text-cyan-400 font-bold mb-4 border-b border-slate-700 pb-2">10's Complement Method</h3>
848     <p class="text-slate-400 mb-2">10's Comp(V2): <span class="text-pink-400">${tens2}</span> (BCD: ${toBCD(tens2)})</p>
849     <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded mb-2"> ${bcd1}<br>+ ${toBCD(tens2)}</p>
850   </div>
851   if (!isNeg) {
852     let final10 = diff10.toString().slice(-maxLen);
853     html += `<p class="text-emerald-400 text-sm mb-2">Carry generated. Discard carry out.</p>
854     <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded">Final: ${toBCD(final10)}</p>`;
855   } else {
856     let tensDiff = diff10.toString().padStart(maxLen, '0');
857     html += `<p class="text-emerald-400 text-sm mb-2">No carry. Result is negative in 10's complement form.</p>
858     <p class="text-white font-bold tracking-widest bg-slate-800 p-2 rounded">Final: ${toBCD(tensDiff)}</p>`;
859   }
860   html += `</div></div>`;
861 }
862 out.innerHTML = html;
863 }
864
865 addCase();
866 addCase();
867 addCase();
868 </script>
869
870 </body>
871 </html>
872
873
874 ### Test Cases
875
876 **Test Case 1: Binary Conversion**
877 I[Sample-1](sample-outputs/Sample-Output-1.png)
878 1. Input Value: 1010
879 2. Selected Base: Binary (Base 2)
```